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Quantifying Distributional Model Risk via Optimal Transport

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Abstract. This paper deals with the problem of quantifying the impact of model misspecification when computing general expected values of interest. The methodology that we propose is applicable in great generality; in particular, we provide examples involving path-dependent expectations of stochastic processes. Our approach consists of computing bounds for the expectation of interest regardless of the probability measure used, as long as the measure lies within a prescribed tolerance measured in terms of a flexible class of distances from a suitable baseline model. These distances, based on optimal transportation between probability measures, include Wasserstein's distances as particular cases. The proposed methodology is well suited for risk analysis and distributionally robust optimization, as we demonstrate with applications. We also discuss how to estimate the tolerance region nonparametrically using Skorokhod-type embeddings in some of these applications.

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Keywords: model risk • distributionally robust optimization using Wasserstein distances • DRO • Kullback–Liebler divergence • ruin probabilities • diffusion approximations

1. Introduction

One of the most ubiquitous applications of probability is the evaluation of performance or risk by means of computations such as $Ef(X) = \int f\,dP$ for a given probability measure P, a function f, and a random element X. In classical applications, such as in finance, insurance, or queueing analysis, X is a stochastic process and f is a path-dependent functional.

The work of the modeler is to choose a probability model P that is both descriptive and yet tractable. However, a substantial challenge, which arises most times, is that the model used, P, will in general be subject to misspecification, either because of a lack of data or by the choice of specific parametric distributions.

The goal of this paper is to develop a framework to assess the impact of model misspecification when computing $Ef(X)$. The importance of constructing systematic methods that provide performance estimates that are robust to model misspecification is emphasized in the academic response to the Basel Accord 3.5, Embrechts et al. [25]: "In its broadest sense, robustness has to do with (in)sensitivity to underlying model deviations and/or data changes. Furthermore, here, a whole new field of research is opening up; at the moment, it is difficult to point to the right approach."

The results of this paper allow the modeler to provide a bound for the expectation of interest regardless of the probability measure used as long as such a measure remains within a prescribed tolerance of a suitable baseline model. The bounds that we obtain are applicable in great generality, allowing us to use our results in settings that include continuous-time stochastic processes and path-dependent expectations in various domains of applications, including insurance and queueing.

Let us describe the approach that we consider more precisely. Our starting point is a technique that has been actively pursued in the literature (to be reviewed momentarily), which involves considering a family of plausible models $\mathcal{P}$ and computing distributionally robust bounds as the solution to the optimization problem,

$$\sup_{P \in \mathcal{P}} \int f\,dP, \tag{1}$$

over all probability measures in $\mathcal{P}$. The motivation, as mentioned earlier, is to measure the highest possible risk regardless of the probability measure in $\mathcal{P}$ used. A canonical object that appears naturally in specifying the family $\mathcal{P}$ is the neighborhood $\{P : d(\mu, P) \le \delta\}$, where μ is a chosen baseline model and δ is a nonnegative tolerance level. Here, d is a metric that measures the discrepancy between probability measures, and the tolerance δ interpolates between no ambiguity ($\delta = 0$) and high levels of model uncertainty (δ large). The family of plausible models, $\mathcal{P}$, can then be specified as a single neighborhood (or a collection of such plausible neighborhoods). In this paper, we choose d in terms of a transport cost (defined precisely in Section 2), and we analyze the solvability of (1) and discuss various implications.

Being able to induce a flexible class of distances that include the popular Wasserstein distances as a special case, transport costs allow for ease of interpretation in terms of the minimum cost associated with transporting mass between probability measures, and they have been widely used in probability theory and its applications (see, e.g., Rachev and Rüschendorf [53, 54], Villani [62], and Ambrosio and Caffarelli [3] for a massive collection of classical applications and Barbour and Xia [8], Gozlan and Leonard [34], Nguyen [50], Wang and Guibas [63], Fournier and Guillin [29], Canas and Rosasco [19], Solomon et al. [60], and Frogner et al. [30] for a sample of a growing list of new applications).

Relative entropy, or Kullback–Liebler (KL) divergence, despite not being a proper metric, has been the most popular choice for d, thanks to the tractability of (1) when d is chosen as relative entropy or other likelihood-based discrepancy measures (see Breuer and Csiszár [18], Lam [45], Chowdhary and Dupuis [20], Glasserman and Xu [32], Atar et al. [5], and Dupuis et al. [23] for the use in robust performance analysis, and see Dupuis et al. [22], Hansen and Sargent [35], Iyengar [41], Nilim and El Ghaoui [51], Lim and Shanthikumar [47], Jain et al. [42], Ben-Tal et al. [11], Wang et al. [64], Jiang and Guan [43], Hu and Hong [38], Bayraksan and Love [10], and references therein for applications to robust control and distributionally robust optimization). Our study is directly motivated to address the shortcoming that many of these earlier works acknowledge: the absolute continuity requirement of relative entropy. There can be two probability measures μ and ν that produce the same samples with high probability, despite having the relative entropy between μ and ν as infinite. Such absolute continuity requirements could be very limiting, particularly so when the models of interest are stochastic processes defined over time. For instance, Brownian motion, because of its tractability in computing path-dependent expectations, is used as approximation of piecewise linear or piecewise constant processes (such as random walk). However, the use of the relative entropy would not be appropriate to accommodate these settings because the likelihood ratio between Brownian motion and any piecewise differentiable process is not well defined.

Another instance that shows the limitations of the KL divergence arises when using an Itô diffusion as a baseline model. In such cases, the region $\mathcal{P}$ contains only Itô diffusions with the same volatility parameter as the underlying baseline model, thus failing to model volatility uncertainty.

The use of a distance d based on an optimal transport cost or Wasserstein's distance, as we do here, also allows for incorporating the use of tractable surrogate models such as Brownian motion. For instance, consider a classical insurance risk problem in which the modeler has sufficient information on claim sizes (e.g., in car insurance) to build a nonparametric reserve model. But the modeler is interested in path-dependent calculations of the reserve process (such as ruin probabilities), so she might decide to use Brownian motion-based approximation as a tractable surrogate model, which allows one to compute $Ef(X)$ easily. A key feature of the Wasserstein distance is that it is computed in terms of a so-called optimal coupling. So the modeler can use any good coupling to provide a valid bound for the tolerance parameter δ.

The use of tractable surrogate models, such as Brownian motion, diffusions, and reflected Brownian motion, has enabled the analysis of otherwise intractable complex stochastic systems. As we shall illustrate, the bounds that we obtain have the benefit of being computable directly in terms of the underlying tractable surrogate model. In addition, the calibration of the tolerance parameter takes advantage of well-studied coupling techniques (such as Skorokhod embedding). So we believe that the approach we propose naturally builds on the knowledge that has been developed by the applied probability community.

We summarize our main contributions in this paper below:

(a) Assuming that X takes values in a Polish space, and using a wide range of optimal transport costs (that include the popular Wasserstein's distances as special cases), we arrive at a dual formulation for the optimization problem in (1) and prove strong duality; see Theorem 1.

(b) Despite the infinite dimensional nature of the optimization problem in (1), we show that the dual problem admits a one-dimensional reformulation that is easy to work with. We provide sufficient conditions for the existence of an optimizer P^* that attains the supremum in (1); see Section 5. Using the result in (a) for upper semicontinuous f, we show how to characterize an optimizer P^* in terms of a coupling involving μ and the chosen transport cost; see Remark 2 and Section 2.4.

(c) We apply our results to various problems such as robust evaluation of ruin probabilities using Brownian motion as a tractable surrogate (see Section 3), general multidimensional first-passage time probabilities (see Section 6.1), and optimal decision making in the presence of model ambiguity (see Section 6.2).

(d) We discuss a nonparametric method, based on a Skorokhod-type embedding, which allows to suitably specify δ (these are particularly useful in the tractable surrogate settings discussed earlier). The specific discussion of the Skorokhod embedding for calibration is given in Appendix A. We also discuss how δ can be chosen in a model misspecification context; see the discussion at the end of Section 6.1.

In a recent paper by Mohajerin Esfahani and Kuhn [49], the authors also consider distributionally robust optimization using a very specific Wasserstein's distance. Similar formulations using Wasserstein distance-based ambiguity sets have been considered in Pflug and Wozabal [52], Wozabal [67], and Zhao and Guan [69] as well. The form of the optimization problem that we consider here is basically the same as that considered in Mohajerin Esfahani and Kuhn [49], but there are important differences that are worth highlighting. In Mohajerin Esfahani and Kuhn [49], the authors concentrate on the specific case of a baseline measure being the empirical distribution of samples obtained from a distribution supported in $\mathbb{R}^d$, and the function f possessing a special structure. By contrast, we allow the baseline measure μ to be supported on general Polish spaces and study a wide class of functions f (upper semicontinuous and integrable with respect to μ). The use of the Skorokhod embeddings that we consider here, for calibration of the feasible region, is also novel and not studied in the present literature.

Another recent paper, of Gao and Kleywegt [31] (note that this is an independent contribution made public a few days after we posted the first version of this paper in the arXiv repository), which presents a very similar version of the strong duality result shown in this paper. One difference that is immediately apparent is that Gao and Kleywegt [31] concentrate on the case in which a specific cost function $c(\cdot,\cdot)$ used to define optimal transport cost is of the form $c(x,y) = d^p(x,y)$ for some $p \geq 1$ and metric $d(\cdot,\cdot)$, whereas we only impose that $c(\cdot,\cdot)$ is lower semicontinuous.

Allowing for general lower semicontinuous cost functions, which may take infinite value or which are nonsymmetric, is useful, as demonstrated in applications such as distributionally robust optimization and machine learning in Blanchet et al. [16, 17] and Sinha et al. [58]. The results in Gao and Kleywegt [31] rely on the existence of limit superior of $|f(y)|/d^p(x,y)$ to be independent of x when x is fixed and $d(x,y) \to \infty$. This feature (i.e., independence of x) is guaranteed if $d(x,y)$ is a metric with compact sublevel sets, but it does not hold for a general (in particular, nonsymmetric) cost function. By contrast, our duality formulation parallels the types of assumptions that are conventional in the theory of optimal transport (see, e.g., theorem 3.1 in Villani [62]).

Another important difference is that, as far as we understand, the proof given in Gao and Kleywegt [31] appears to implicitly assume that the space S in which the random element of interest X takes values is locally compact. For example, the proof of lemma 2 in Gao and Kleywegt [31] and other portions of the technical development appear to assume repeatedly that a closed norm ball is compact, a property that does not hold in any infinite-dimensional topological vector space.[1]

Under different assumptions on the transport cost function, the strong duality result we derive is also of interest in a much more recent work, Bartl et al. [9], devoted to robust optimized certainty equivalents.

The rest of the paper is organized as follows: In Section 2 we shall introduce the assumptions and discuss our main result, whose proof is given in Section 4. The proof is technical because of the level of generality that is considered. More precisely, it is the fact that the cost functions used to define optimal transport costs need not be continuous, and the random elements that we consider need not take values in a locally compact space (such as $\mathbb{R}^d$), which introduces technical complications, including issues such as the measurability of a key functional appearing in the dual formulation. In preparation for the proof, and to help the reader gain some intuition of the result, we present a one-dimensional example first, in Section 3. Then, after providing the proof of our main result in Section 4 and conditions for the existence of the worst-case probability distribution that attains the supremum in (1) in Section 5, we discuss additional examples in Section 6.

2. Our Main Result

To state our main result, we need to introduce some notation and define the optimal transport cost between probability measures.

2.1. Notation and Definitions

For a given Polish space S, we use $\mathcal{B}(S)$ to denote the associated Borel σ-algebra. Let us write $P(S)$ and $M(S)$ to denote the set of all probability measures and finite signed measures on $(S, \mathcal{B}(S))$, respectively. For any $\mu \in P(S)$, let $\mathcal{B}_\mu(S)$ denote the completion of $\mathcal{B}(S)$ with respect to μ; the unique extension of μ to $\mathcal{B}_\mu(S)$ that is a probability measure is also denoted by μ, and the measure μ in $\int \varphi d\mu$ is to be interpreted as this extension defined on $\mathcal{B}_\mu(S)$ whenever $\varphi : (S, \mathcal{B}(S)) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is not measurable but, instead, $\varphi : (S, \mathcal{B}_\mu(S)) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is measurable. We shall use $C_b(S)$ to denote the space of bounded continuous functions from S to $\mathbb{R}$ and $\mathrm{Spt}(\mu)$

to denote the support of a probability measure μ. We say that μ is *concentrated* on a set $A \in B_\mu(S)$ if $\mu(A) = 1$. For any $\mu \in P(S)$ and $p \geq 1, L^p(d\mu)$ denotes the collection of Borel-measurable functions $h: S \to \mathbb{R}$ such that $\int |h|^p d\mu < \infty$. The universal σ–algebra is defined by $\mathcal{U}(S) = \cap_{\mu \in P(S)} B_\mu(S)$. We use $\mathfrak{m}_{\mathcal{U}}(S; \mathbb{R} \cup \{+\infty\})$ to denote the collection of measurable functions $\varphi: (S, \mathcal{U}(S)) \to (\mathbb{R} \cup \{+\infty\}, \mathcal{B}(\mathbb{R} \cup \{+\infty\}))$. As $\mathcal{U}(S) \subseteq \mathcal{B}_\mu(S)$ for every $\mu \in P(S)$, any $\varphi \in \mathfrak{m}_{\mathcal{U}}(S; \mathbb{R} \cup \{+\infty\})$ is also measurable when S and $\mathbb{R} \cup \{+\infty\}$ are equipped with the σ-algebras $\mathcal{B}_\mu(S)$ and $\mathcal{B}(\mathbb{R} \cup \{+\infty\})$, respectively. Consequently, the integral $\int \varphi d\mu$ is well defined for any nonnegative $\varphi \in \mathfrak{m}_{\mathcal{U}}(S; \mathbb{R} \cup \{+\infty\})$.

2.1.1. Optimal Transport Cost. For any two probability measures μ_1 and μ_2 in $P(S)$, let $\Pi(\mu_1, \mu_2)$ denote the set of all joint distributions with μ_1 and μ_2 as respective marginals. In other words, the set $\Pi(\mu_1, \mu_2)$ represents the set of all couplings (also called transport plans) between μ_1 and μ_2. Throughout the paper, we assume that the following.

Assumption 1 (A1). The function c : $c: S \times S \to [0, +\infty]$ *is a lower semicontinuous function satisfying* $c(x, x) = 0$ *for every* $x \in S$.

Then the *optimal transport cost* associated with the cost function c is defined as

$$d_c(\mu_1, \mu_2) := \inf\left\{\int c d\pi : \pi \in \Pi(\mu_1, \mu_2)\right\}, \quad \mu_1, \mu_2 \in P(S). \tag{2}$$

Intuitively, the quantity $c(x, y)$ specifies the cost of transporting unit mass from x in S to another element y of S. Given a lower semicontinuous cost function $c(x, y)$ and a coupling $\pi \in \Pi(\mu_1, \mu_2)$, the integral $\int c d\pi$ represents the expected cost associated with the coupling (or transport plan) π. For any nonnegative lower semicontinuous cost function c, it is known that the *optimal transport plan* that attains the infimum in the above definition exists (see theorem 4.1 of Villani [62]), and therefore, the optimal transport cost $d_c(\mu_1, \mu_2)$ corresponds to the lowest transport cost that is attainable among all couplings between μ_1 and μ_2.

If the cost function c is symmetric (i.e., $c(y, x) = c(x, y)$ for all x, y), is real-valued, and satisfies triangle inequality, one can use the minimum transportation cost $d_c(\cdot, \cdot)$ to define a distance on the space of probability measures. For example, if S is a Polish space equipped with metric d, then taking the cost function to be $c(x, y) = d^p(x, y)$, for some $p \in [1, \infty)$, renders $d_c^{1/p}(\mu, \nu)$ to be the well-known Wasserstein distance order p between μ and ν. Unlike the Kullback–Liebler divergence or other likelihood-based divergence measures, the Wasserstein distances can be shown to satisfy the axioms of distance. More importantly, Wasserstein distances do not restrict all the probability measures in the neighborhoods such as $\{\nu \in P(S) : d_c(\mu, \nu) \leq \delta\}$ to share the same support as that of μ (see, e.g., chapter 6 in Villani [62] for properties of Wasserstein distances).

2.2. Primal Problem

Underlying our discussion, we have a Polish space S, which is the space where the random elements of the given probability model $\mu \in P(S)$ take values. Given $\delta \in (0, +\infty)$, the objective, as mentioned in the introduction, is to evaluate

$$\sup\left\{\int f d\nu : d_c(\mu, \nu) \leq \delta\right\},$$

for any function f that satisfies the following assumption.

Assumption 2 (A2). The function f is $f: S \to \mathbb{R}$ *is upper semicontinuous, and* $f \in L^1(d\mu)$.

As the integral $\int f d\nu$ may equal $+\infty - \infty$ for some ν satisfying $d_c(\mu, \nu) \leq \delta$, only for the purposes of interpreting the supremum above, we define $+\infty - \infty := +\infty$.[2] The value of this optimization problem provides a bound to the expectation of f, regardless of the probability measure used, as long as the measure lies within δ distance (measured in terms of d_c) from the baseline probability measure μ. In applied settings, μ is the probability measure chosen by the modeler as the baseline distribution, and the function f corresponds to a risk functional or performance measure of interest—for example, expected losses, probability of ruin, and so forth.

As the infimum in (2) is attained for any given nonnegative lower semicontinuous cost function c (see theorem 4.1 of Villani [62]), we rewrite the quantity of interest as follows:

$$\bar{I} := \sup\left\{\int f d\nu : d_c(\mu, \nu) \leq \delta\right\} = \sup\left\{\int f(y) d\pi(x, y) : \pi \in \bigcup_{\nu \in P(S)} \Pi(\mu, \nu), \int c d\pi \leq \delta\right\},$$

which, in turn, is an optimization problem with a linear objective function and linear constraints in terms of the variable π. If we let

$$I(\pi) := \int f(y) d\pi(x,y) \quad \text{and} \quad \Phi_{\mu,\delta} := \left\{ \pi \in \bigcup_{\nu \in P(S)} \Pi(\mu,\nu) : \int c d\pi \le \delta \right\}$$

for brevity, then

$$\bar{I} = \sup \{ I(\pi) : \pi \in \Phi_{\mu,\delta} \} \tag{3}$$

is the quantity of interest. Since $d_c(\mu,\mu) = 0$, the set $\Phi_{\mu,\delta}$ is not empty for any $\delta > 0$.

2.2.1. The Dual Problem and Weak Duality. Let $C := \{(x,y) \in S \times S : c(x,y) < \infty\}$. Since $c(\cdot,\cdot)$ is lower semi-continuous, $C \in \mathcal{B}(S \times S)$. Define $\Lambda_{c,f}$ to be the collection of pairs (λ, φ) such that λ is a nonnegative real number, $\varphi \in \mathfrak{m}_{\mathcal{U}}(S; \mathbb{R} \cup \{+\infty\})$, and

$$\varphi(x) + \lambda c(x,y) \ge f(y), \ \text{ for all } (x,y) \in C. \tag{4}$$

For every such $(\lambda,\varphi) \in \Lambda_{c,f}$, consider

$$J(\lambda,\varphi) := \lambda\delta + \int \varphi d\mu.$$

As $\int f\, d\mu$ is finite and $\varphi \ge f$ whenever φ satisfies (4), the integral in the definition of $J(\lambda,\varphi)$ avoids ambiguities such as $\infty - \infty$ for any $(\lambda,\varphi) \in \Lambda_{c,f}$. Furthermore, for any $\pi \in \Phi_{\mu,\delta}$, as $\int c d\pi$ is finite, we have $\pi(C) = 1$ and $\int_C g d\pi = \int g d\pi$ for every measurable $g : (S \times S, \mathcal{B}(S \times S)) \to (\mathbb{R} \cup \{+\infty\}, \mathcal{B}(\mathbb{R} \cup \{+\infty\}))$. Therefore, for any $\pi \in \Phi_{\mu,\delta}$ and $(\lambda,\varphi) \in \Lambda_{c,f}$, see that

$$\begin{aligned} J(\lambda,\varphi) &= \lambda\delta + \int_C \varphi(x) d\pi(x,y) \ge \lambda\delta + \int_C (f(y) - \lambda c(x,y))\, d\pi(x,y) \\ &= \lambda\delta + \int (f(y) - \lambda c(x,y)) d\pi(x,y) \\ &= \int f(y) d\pi(x,y) + \lambda\left(\delta - \int c(x,y) d\pi(x,y)\right) \\ &\ge \int f(y) d\pi(x,y) \\ &= I(\pi). \end{aligned}$$

Consequently, we have

$$\underline{J} := \inf \{ J(\lambda,\varphi) : (\lambda,\varphi) \in \Lambda_{c,f} \} \ge \bar{I}. \tag{5}$$

Following the tradition in optimization theory, we refer to the above infimum problem that solves for $\underline{J}$ in (5) as the dual to the problem that solves for $\bar{I}$ in (3), which we address as the primal problem. Our objective in the next section is to identify whether the primal and the dual problems have same value (i.e., do we have that $\bar{I} = \underline{J}$?).

2.3. Main Result: Strong Duality Holds

Recall that the feasible sets for the primal and dual problems, respectively, are

$$\Phi_{\mu,\delta} := \left\{ \pi \in \bigcup_{\nu \in P(S)} \Pi(\mu,\nu) : \int c d\pi \le \delta \right\} \quad \text{and} \tag{6a}$$

$$\Lambda_{c,f} := \{ (\lambda,\varphi) : \lambda \ge 0,\ \varphi \in \mathfrak{m}_{\mathcal{U}}(S; \mathbb{R} \cup \{+\infty\}),\ \varphi(x) + \lambda c(x,y) \ge f(y) \ \text{ for all } (x,y) \in C \}. \tag{6b}$$

The corresponding primal and dual problems are

$$\bar{I} := \sup\left\{ \int f(y) d\pi(x,y) : \pi \in \Phi_{\mu,\delta} \right\} \quad \text{and} \quad \underline{J} := \inf\left\{ \lambda\delta + \int \varphi d\mu : (\lambda,\varphi) \in \Lambda_{c,f} \right\}.$$

For brevity, we have identified the primal and dual objective functions as $I(\pi)$ and $J(\lambda,\varphi)$, respectively. As the identified primal and dual problems are infinite dimensional, it is not immediate whether they have same value (i.e., is $\bar{I} = \underline{J}$?). The objective of Theorem 1 is to verify that, for a broad class of performance measures f, $\bar{I}$ indeed equals $\underline{J}$.

Before stating Theorem 1, for every $\lambda \ge 0$, we define $\varphi_\lambda : S \to \mathbb{R}\cup\{+\infty\}$ as follows:

$$\varphi_\lambda(x) := \begin{cases} \sup_{y\in S}\{f(y) - \lambda c(x,y)\} & \text{if } \lambda > 0, \\ \sup\{f(y) : c(x,y) < \infty\} & \text{if } \lambda = 0. \end{cases} \tag{8}$$

For notational convenience, *we write $\lambda c(x,y) = +\infty$ whenever $\lambda = 0$ and $c(x,y) = +\infty$ throughout this paper*; this allows us to reexpress $\varphi_\lambda(\cdot)$ conveniently as

$$\varphi_\lambda(x) := \sup_{y\in S}\{f(y) - \lambda c(x,y)\}, \quad x \in S, \tag{9}$$

without having to write in cases (as in (8)).

Theorem 1 (Strong Duality Holds). *Under the Assumptions* (A1) *and* (A2),
(a) $\bar{I} = \underline{J}$. *In other words,*

$$\sup\{I(\pi) : \pi \in \Phi_{\mu,\delta}\} = \inf\{J(\lambda,\varphi) : (\lambda,\varphi) \in \Lambda_{c,f}\}.$$

(b) *There exists a dual optimizer of the form $(\lambda,\varphi_\lambda)$ for some $\lambda \ge 0$ and $\varphi_\lambda(\cdot)$ defined as in* (8) (*or, equivalently,* (9)). *In addition, any feasible $\pi^* \in \Phi_{\mu,\delta}$ and $(\lambda^*,\varphi_{\lambda^*}) \in \Lambda_{c,f}$ are primal and dual optimizers, satisfying $I(\pi^*) = J(\lambda^*,\varphi_{\lambda^*})$, if and only if*

$$f(y) - \lambda^* c(x,y) = \sup_{z\in S}\{f(z) - \lambda^* c(x,z)\}, \quad \pi^* \text{ a.s., and} \tag{10a}$$

$$\lambda^*\left(\int c(x,y)\,d\pi^*(x,y) - \delta\right) = 0. \tag{10b}$$

Moreover, the primal optimizer π^*, if it exists, is unique if for μ-almost every $x \in S$, there is only one y in S that attains the supremum in $\sup_{y\in S}\{f(y) - \lambda^* c(x,y)\}$.

As the measurability of the function φ_λ is not immediate, we establish that $\varphi_\lambda \in \mathfrak{m}_{\mathcal{U}}(S;\mathbb{R}\cup\{+\infty\})$ in Section 4, where the proof of Theorem 1 is also presented. Sufficient conditions for the existence of a primal optimizer $\pi^* \in \Phi_{\mu,\delta}$, satisfying $I(\pi^*) = \bar{I}$, are presented in Section 5. For now, we are content discussing the insights that can be obtained from Theorem 1.

Remark 1 (On the Value of the Dual Problem). First, we point out the following useful characterization of the optimal value, $\bar{I}$, as a consequence of Theorem 1:

$$\bar{I} = \inf_{\lambda\ge 0}\left\{\lambda\delta + E_\mu\left[\sup_{y\in S}\{f(y) - \lambda c(X,y)\}\right]\right\}, \tag{11}$$

where the right-hand side[3] is simply a univariate reformulation of the dual problem, with $\lambda c(x,y)$ taken as $+\infty$ whenever $\lambda = 0$ and $c(x,y) = +\infty$ (see (8) and (9)). This characterization of $\bar{I}$ follows from the strong duality in Theorem 1 and the observation that $J(\lambda,\varphi_\lambda) \le J(\lambda,\varphi)$ for every $(\lambda,\varphi) \in \Lambda_{c,f}$. The significance of this observation lies in the fact that the only probability measure involved in the right-hand side of (11) is the baseline measure μ, which is completely characterized and is usually chosen in a way that it is easy to work with or draw samples from. In effect, the result indicates that the infinite-dimensional optimization problem in (3) can as well be solved by working with the univariate reformulation in the right-hand side of (11).

Remark 2 (On the Structure of Primal Optimal Transport Plan).[4] It is evident from the characterization of an optimal measure π^* in Theorem 1(b) that if π^* exists, it is concentrated on $\{(x,y) \in S\times S : y \in \arg\max_{z\in S}\{f(z) - \lambda^* c(x,z)\}\}$. Thus, a worst-case joint probability measure π^* gets identified with a transport plan that transports mass from x to the optimizer(s) of the local optimization problem $\sup_{z\in S}\{f(z) - \lambda^* c(x,z)\}$. According to the complementary slackness conditions (10a) and (10b), such a transport plan π^* satisfies one of the following two cases.

Case 1. When $\lambda^* > 0$: The transport plan π^* necessarily costs $\int c d\pi^* = \delta$.

Case 2. When $\lambda^* = 0$: The transport plan π^* satisfies $\int c d\pi^* \leq \delta$ (follows from primal feasibility of π^*) and $f(y) = \sup\{f(z) : c(x,z) < \infty\}$, π^* almost surely (follows from (10a), assuming that $\arg\max\{f(z) : c(x,z) < \infty\}$ exists). The interpretation is that the budget for quantifying ambiguity, δ, is sufficiently large to move all the probability mass to $\arg\max\{f(z) : c(x,z) < \infty\}$, thus making $\bar{I}$ as large as possible.

Remark 3. If $\sup_{y \in S} f(y)/(1 + c(x,y)) = \infty$ (e.g., if $f(y)$ grows to ∞ at a rate faster than the rate at which the transport cost function $c(x,y)$ grows) for every x in a set $A \subseteq S$ such that $\mu(A) > 0$, then $\bar{I} = \underline{J} = \infty$. This is because, for every $\lambda \geq 0$ and $x \in A$, $\varphi_\lambda(x) = \sup_{y \in S}\{f(y) - \lambda c(x,y)\} = \infty$, and consequently, $\lambda\delta + \int \varphi_\lambda d\mu = \infty$ for every $\lambda \geq 0$; therefore, $\bar{I} = \underline{J} = \infty$ as a consequence of Theorem 1. Requiring the objective f to not grow faster than the transport cost c may be useful from a modeling viewpoint as it offers guidance in understanding choices of transport cost functions that necessarily yield $\sup\{\int f dv : d_c(\mu, v) \leq \delta\} = \infty$.

The following corollary of Theorem 1 provides a characterization of ε-optimal transport plans.

Corollary 1. *Suppose that Assumptions (A1) and (A2) are in force, and $(\lambda^*, \varphi_{\lambda^*}) \in \Lambda_{c,f}$ is dual optimal (i.e., $J(\lambda^*, \varphi_{\lambda^*}) = \underline{J}$. Then for any $\varepsilon > 0$, $\pi_\varepsilon \in \Phi_{\mu,\delta}$ is ε-primal optimal (i.e., $I(\pi_\varepsilon) \geq J(\lambda^*, \varphi_{\lambda^*}) - \varepsilon$) if and only if*

$$\int (\varphi_{\lambda^*}(x) - (f(y) - \lambda^* c(x,y))\, d\pi_\varepsilon(x,y) + \lambda^* \left(\delta - \int c d\pi_\varepsilon\right) \leq \varepsilon. \tag{12}$$

Proof. The proof of Corollary 1 follows trivially by substituting the following expressions for $\bar{I}$ and $I(\pi_\varepsilon)$ in $\bar{I} - I(\pi_\varepsilon) \leq \varepsilon$.

$$\bar{I} = J(\lambda^*, \varphi_{\lambda^*}) = \lambda^*\delta + \int \varphi_{\lambda^*}(x) d\pi_\varepsilon(x,y) \text{ and}$$

$$I(\pi_\varepsilon) = \int f(y)\, d\pi_\varepsilon(x,y) = \int (f(y) - \lambda^* c(x,y))\, d\pi_\varepsilon(x,y) + \lambda^* \int c(x,y) d\pi_\varepsilon(x,y). \quad \square$$

As both the summands in (12) are nonnegative, for any $\pi_\varepsilon \in \Phi_{\mu,\delta}$ such that $I(\pi_\varepsilon) \geq \bar{I} - \varepsilon$, we have

$$\int (\varphi_{\lambda^*}(x) - f(y) - \lambda^* c(x,y))\, d\pi_\varepsilon(x,y) \leq \varepsilon \quad \text{and} \quad \left(\delta - \frac{\varepsilon}{\lambda^*}\right)^+ \leq \int c d\pi_\varepsilon \leq \delta, \tag{13}$$

where $a^+ := \max\{a, 0\}$ for any $a \in \mathbb{R}$.

We next discuss an important special case of Theorem 1—namely, computing worst-case probabilities. The applications of this special case, in the context of ruin probabilities, are presented in Sections 3 and 6.1. Other applications of Theorem 1—broadly, in the context of distributionally robust optimization—are available in Mohajerin Esfahani and Kuhn [49], Zhao and Guan [69], Gao and Kleywegt [31], and Blanchet et al. [16], as well in Example 4 in Section 6.2 of this paper.

2.4. Application of Theorem 1 for Computing Worst-Case Probabilities

Suppose that we are interested in computing

$$\bar{I} = \sup\{P(A) : d_c(\mu, P) \leq \delta\}, \tag{14}$$

where A is a nonempty closed subset of the Polish space S. Since A is closed, the function $f(x) = \mathbf{1}_A(x)$ is upper semicontinuous, and therefore we can apply Theorem 1 to address (14). To apply Theorem 1, we first observe that $\sup_{y \in S}\{\mathbf{1}_A(y) - \lambda c(x,y)\} = (1 - \lambda c(x,A))^+$, where $c(x,A) := \inf\{c(x,y) : y \in A\}$ is the lowest cost possible in transporting unit mass from $x \in S$ to some y in A, and a^+ denotes the positive part of the real number a; here, since $\lambda c(x,y) = +\infty$ whenever $\lambda = 0$ and $c(x,y) = +\infty$, we have $\lambda c(x,A) = +\infty$ whenever $\lambda = 0$ and $c(x,A) = +\infty$. Consequently, Theorem 1 guarantees that the quantity of interest, $\bar{I}$, in (14) can be computed by solving

$$\bar{I} = \inf_{\lambda \geq 0}\left\{\lambda\delta + E_\mu\left[(1 - \lambda c(X,A))^+\right]\right\}. \tag{15}$$

Theorem 2 presents a further useful characterization of $\bar{I} = \sup\{P(A) : d_c(\mu, P) \leq \delta\}$ and also sheds some light on the structure of the primal optimal transport plan. To state Theorem 2, we need the following definitions: Let $h : \mathbb{R}_+ \cup \{+\infty\} \to \mathbb{R}_+ \cup \{+\infty\}$ be defined as

$$h(u) := \int_{\{x:c(x,A)\leq u\}} c(x, A) d\mu(x). \tag{16}$$

Since $c(\cdot, \cdot)$ is nonnegative, $h(\cdot)$ is nondecreasing; furthermore, it follows from the monotone convergence theorem that $h(\cdot)$ is a right continuous function with left limits equal to $\int_{\{x:c(x,A)<u\}} c(x, A) d\mu(x)$. We identify its generalized inverse $h^{-1} : \mathbb{R}_+ \cup \{+\infty\} \to \mathbb{R}_+ \cup \{+\infty\}$ as $h^{-1}(t) := \inf\{u \geq 0 : h(u) \geq t\}$. Next, define

$$\underline{c} := \int_{\{x:c(x,A)<h^{-1}(\delta)\}} c(x, A) d\mu(x) \quad \text{and} \quad \bar{c} := \int_{\{x:c(x,A)\leq h^{-1}(\delta)\}} c(x, A) d\mu(x). \tag{17}$$

With the above definitions, observe that $\underline{c} \leq \delta \leq \bar{c} = h(h^{-1}(\delta))$, and $\underline{c} = \bar{c} = \delta$ if $\mu(x \in S : c(x, A) = h^{-1}(\delta)) = 0$. Along with these definitions, let us recall our notational conventions that $1/0 = +\infty, \inf \varnothing = +\infty$ and $\lambda c(x, y) = +\infty$ whenever $\lambda = 0$ and $c(x, y) = +\infty$ (and likewise, $\lambda c(x, A) = +\infty$ whenever $\lambda = 0$ and $c(x, A) = +\infty$).

Theorem 2. *Suppose that Assumption (A1) is in force, and A is a nonempty closed subset of the Polish space S. Then, for any* $\delta \in (0, \infty)$,

(a) *the choice* $\lambda = 1/h^{-1}(\delta)$ *attains the infimum in* (15);

(b) *if* $\mu(x \in S : c(x, A) = h^{-1}(\delta)) = 0$, *then* $\bar{I} = \mu(x \in S : c(x, A) < h^{-1}(\delta))$;

(c) *if* $\mu(x \in S : c(x, A) = h^{-1}(\delta)) \neq 0$, *then* $\bar{I} = \mu(x : c(x, A) < h^{-1}(\delta)) + \frac{\delta - \underline{c}}{\bar{c} - \underline{c}} \mu(x : c(x, A) = h^{-1}(\delta))$, *and*

(d) *if a primal optimal transport plan* $\pi^* \in \Phi_{\mu,\delta}$ *satisfying* $I(\pi^*) := \pi^*(S \times A) = \bar{I}$ *exists, it is concentrated on* $S^* := \{(x, y) : c(x, A) \leq h^{-1}(\delta),\ y \in Proj_A(x)\} \cup \{(x, x) : c(x, A) \geq h^{-1}(\delta)\}$, *where* $Proj_A(x) := \{y \in A : c(x, y) = c(x, A)\}$.

2.4.1. Proof of Theorem 2. Recall our notational convention that $1/0 = +\infty, \inf \varnothing = +\infty$ and $\lambda c(x, y) = +\infty$ whenever $\lambda = 0$ and $c(x, y) = +\infty$ (and likewise, $\lambda c(x, A) = +\infty$ whenever $\lambda = 0$ and $c(x, A) = +\infty$). The proof utilizes Lemmas 1 and 2, which are proved in Technical Appendix B.

Lemma 1. *Suppose that Assumption (A1) is in force, and A is a nonempty closed subset of the Polish space S. For any* $\lambda > 0$, *if* $(\pi_n : n \geq 1) \subseteq \cup_{\nu \in P(S)} \Pi(\mu, \nu)$ *is such that*

$$\int ((1 - \lambda c(x, A))^+ - (1_A(y) - \lambda c(x, y))) d\pi_n(x, y) < \frac{1}{n}, \tag{18}$$

then $\lim_{v \uparrow 1/\lambda} h(v) \leq \underline{\lim}_n \int c d\pi_n \leq \overline{\lim}_n \int c d\pi_n \leq h(1/\lambda)$.

Lemma 2. *Suppose that Assumption (A1) is in force, and A is a nonempty closed subset of the Polish space S. If* $(\pi_n : n \geq 1) \subseteq \cup_{\nu \in P(S)} \Pi(\mu, \nu)$ *is such that* (18) *holds for* $\lambda = 0$, *then* $\underline{\lim}_n \int c d\pi_n \geq \lim_{v \uparrow \infty} h(v)$.

Proof of Theorem 2(a). For any $\lambda \geq 0$, if we let $f(x) = \mathbf{1}_A(x)$, then $\varphi_\lambda(x) := \sup\{f(y) - \lambda c(x, y)\} = (1 - \lambda c(x, A))^+$. Suppose that $\lambda^* \in [0, \infty)$ attains the infimum in (15); the existence of such a λ^* follows from Theorem 1(b). Consequently, it follows from Corollary 1 that there exists a sequence of probability measures $(\pi_n : n \geq 1) \subseteq \Phi_{\mu,\delta}$ such that

$$\int ((1 - \lambda^* c(x, A))^+ - (1_A(y) - \lambda^* c(x, y))) \, d\pi_n(x, y) \leq \frac{1}{n} \quad \text{and} \quad \lambda^* \left(\delta - \int c d\pi_n\right) \leq \frac{1}{n}. \tag{19}$$

Since $(\pi_n : n \geq 1)$ satisfies (18), we obtain from Lemmas 1 and 2 that $\underline{\lim}_n \int c d\pi_n \geq \lim_{v \uparrow 1/\lambda^*} h(v)$ when $\lambda^* \geq 0$ and $\overline{\lim}_n \int c d\pi_n \leq h(1/\lambda^*)$ when $\lambda^* > 0$. Furthermore, as $\int c d\pi_n \leq \delta$ for all n, it follows from the deduction $\underline{\lim}_n \int c d\pi_n \geq \lim_{v \uparrow 1/\lambda^*} h(v)$ that $\lim_{v \uparrow 1/\lambda^*} h(v) \leq \delta$.

If $\lambda^* = 0$, it is immediate from $\lim_{v \uparrow \infty} h(v) \leq \delta$ that $h^{-1}(\delta) = \infty$, and consequently, $1/h^{-1}(\delta) = 0 = \lambda^*$ is an optimal choice of λ in (15). Therefore, it remains to argue that the choice $\lambda = 1/h^{-1}(\delta)$ attains the infimum in (15) if λ^* is strictly positive.

If $\lambda^* > 0$, it follows from the second inequality in (19) and the earlier deduction $\overline{\lim}_n \int c d\pi_n \leq h(1/\lambda^*)$ that $h(1/\lambda^*) \geq \delta$. Since $\lim_{v \uparrow 1/\lambda^*} h(v) \leq \delta \leq h(1/\lambda^*)$, at least one of the following is true: (a) $1/\lambda^* = h^{-1}(\delta)$, or (b) $\lambda^* \in \Lambda := \{\lambda > 0 : h(1/\lambda) = \delta\}$. If Λ is empty, then λ^* must necessarily equal $1/h^{-1}(\delta)$, which proves the claim in

Theorem 2(a). As it remains to check the claim only when Λ is not empty, we assume that $\lambda^* > 0$ and $\Lambda \neq \varnothing$ for the rest of the proof. In this case, observe that $h^{-1}(\delta) \in \Lambda$. In addition, for any $\lambda \in \Lambda$,

$$\begin{aligned}\lambda\delta + E_\mu\big[(1-\lambda c(X,A))^+\big] &= \lambda(\delta - E_\mu[c(X,A); c(X,A) \leq 1/\lambda]) + \mu(x \in S : \mathrm{c}(x,A) \leq 1/\lambda)\\ &= \lambda(\delta - h(1/\lambda)) + \mu(x \in S : c(x,A) \leq 1/\lambda)\end{aligned}$$

which, in turn, equals $\mu(x \in S : c(x,A) \leq 1/h^{-1}(\delta))$; this is because $h(1/\lambda) = \delta$ for any $\lambda \in \Lambda$, and consequently, $\mu(x \in S : c(x,A) \in (\inf_{\lambda\in\Lambda} 1/\lambda, \sup_{\lambda\in\Lambda} 1/\lambda)) = 0$. Since the value of the objective function, $\lambda\delta + E_\mu[(1-\lambda c(X,A))^+]$, is identical for any $\lambda \in \Lambda \supseteq \{1/h^{-1}(\delta)\}$, it follows that the choice $\lambda = 1/h^{-1}(\delta)$ as well attains the infimum in (15).

Without loss of generality, we take $\lambda^* = 1/h^{-1}(\delta)$ for the rest of the proof of Theorem 2.

Proof of Theorem 2(b). Since the infimum in (15) is attained at $\lambda^* = 1/h^{-1}(\delta)$, we have

$$\bar{I} = \lambda^*\delta + E_\mu\big[(1-\lambda c(X,A))^+\big] = \lambda^*\big(\delta - E_\mu\big[c(X,A); c(X,A) < 1/\lambda^*\big]\big) + \mu\big(x \in S : c(X,A) < 1/\lambda^*\big). \tag{20}$$

Since $\mu(x : c(x,A) = h^{-1}(\delta)) = 0$, we have $\bar{c} = \delta = \underline{c} := E_\mu[c(X,A); c(X,A) < 1/\lambda^*]$ whenever $\lambda^* > 0$. Therefore, it is immediate from (20) that $\bar{I} = \mu(x \in S : c(X,A) < 1/\lambda^*)$.

Proof of Theorem 2(c). Since $E_\mu[c(X,A); c(X,A) \leq h^{-1}(\delta)] = E_\mu[c(X,A); c(X,A) < h^{-1}(\delta)] + h^{-1}(\delta)\mu(x : c(x,A) = h^{-1}(\delta))$, we have $h^{-1}(\delta) = (\bar{c} - \underline{c})/\mu(x \in S : c(x,A) = h^{-1}(\delta))$. Substituting $\lambda^* = 1/h^{-1}(\delta)$ in (20), we establish the claim in Theorem 2(c).

Proof of Theorem 2(d). It follows from Theorem 2(b) that if a primal optimizer π^* satisfying $\bar{I} = \pi^*(S \times A)$ exists, it is concentrated on $\{(x,y) \in S \times S : y \in \arg\max_{z\in S}\{f(z) - \lambda^* c(x,z)\}\}$. Now, the claim in Theorem 2(d) is immediate from the following observation:

$$\arg\max_{y\in S}\{\mathbf{1}_A(y) - \lambda^* c(x,y)\} = \begin{cases} Proj_A(x) & \text{if } 0 \leq \lambda^* c(x,A) < 1,\\ Proj_A(x) \cup \{x\} & \text{if } \lambda^* c(x,A) = 1\\ \{x\} & \text{otherwise,}\end{cases} \tag{21}$$

2.4.2. A Discussion on Theorem 2. The reformulation that $\bar{I} = \mu(x : c(x,A) < 1/\lambda^*)$, as in Theorem 2(b), is useful because it reexpresses the worst-case probability of interest in terms of the probability of a suitably inflated neighborhood, $\{x \in S : c(x,A) \leq 1/\lambda^*\}$, evaluated under the reference measure μ, which is often tractable (see Sections 3 and 6.1 for some applications).

For instance, if we take the cost function c as a distance metric d (as in Wasserstein distances), and let the baseline measure μ and the set A be such that the function $h(\cdot)$, which is defined in (16), is continuous. Then Theorem 2(b) allows us to write the worst-case probability $\sup\{P(A) : d_c(\mu,P) \leq \delta\}$ simply as $\mu(A_{1/\lambda^*})$, where for any $\varepsilon > 0$, $A_\varepsilon := \{y \in S : d(x,y) \leq \varepsilon \text{ for some } x \in A\}$ denotes the ε-neighborhood of the set A, and λ^* is the solution to the univariate optimization problem in (15). In addition, because of the characterization of the optimal lambda in Theorem 2(a), we have $1/\lambda^* = h^{-1}(\delta) := \inf\{u \geq 0 : h(u) \geq \delta\}$. If $h(u)$ does not admit a closed-form expression, one approach is to obtain samples of X under the reference measure μ and either solve the sampled version of (15) or compute a Monte Carlo approximation of the integral $h(u)$ to identify the level $1/\lambda^*$ as $h^{-1}(\delta) = \inf\{u : h(u) \geq \delta\}$. Next, Lemma 3 discusses the structure of optimal transport plans.

Lemma 3. *Suppose that Assumptions (A1) and (A2) are in force, the set A is a nonempty closed subset of S, and $Proj_A(x) \neq \varnothing$[5] for μ-almost every x. In addition, suppose that $\delta \in (0,\infty)$ is such that $\lambda^* := 1/h^{-1}(\delta) \in (0,\infty)$. Let X be distributed according to μ, and let Z be an independent Bernoulli random variable with success probability*

$$p := \begin{cases} 0 & \textit{if } \underline{c} = \bar{c},\\ (\delta - \underline{c})/(\bar{c} - \underline{c}) & \textit{otherwise,}\end{cases}$$

where the quantities $\underline{c}, \bar{c}$, with $\delta \in [\underline{c}, \bar{c}]$, are defined as in (17). Then,

(a) *there exists a universally measurable map $\xi : S \to A$ such that $\xi(x) \in Proj_A(x)$, μ–almost surely; and*
(b) *if we let*

$$Y^* := \begin{cases} \xi(X) & \textit{if } c(X,A) < 1/\lambda^* \textit{ and } Z = 1,\\ X & \textit{otherwise,}\end{cases}$$

then the law of (X, Y^*) attains the supremum in $\bar{I} = \sup\{\pi(S \times A) : \pi \in \Phi_{\mu,\delta}\}$.

Proof of Lemma 3. Since $c(\cdot,\cdot)$ is lower semicontinuous and $Proj_A(x)$ is not empty almost surely, the existence of $\xi : S \to A$ with desired properties follows from standard measurable selection results (see, e.g., proposition 7.50(b) of Bertsekas and Shreve [13]).

To prove part (b), let $\Pi_{S^*}(\mu)$ be the collection of probability measures in $\cup_{\nu \in P(S)} \Pi(\mu,\nu)$ satisfying $\pi(S^*) = 1$ (see Theorem 2(d) for the definition of S^*). If $\lambda^* = 1/h^{-1}(\delta)$ is positive and $\underline{c} = \bar{c} = \delta$, any coupling in $\Pi_{S^*}(\mu)$ is primal optimal (because it satisfies the complementary slackness conditions (10a) and (10b) in addition to the primal feasibility condition that $\pi^* \in \Phi_{\mu,\delta}$). In this case, $p = 0$, and the described coupling $(X, Y^*) \in \Pi_{S^*}(\mu)$. Therefore, the law of (X, Y^*) is primal optimal.

On the other hand, if $\lambda^* = 1/h^{-1}(\delta) \in (0,\infty)$ is such that $\underline{c} < \bar{c}$, the following hold: $\Pr((X, Y^*) \in S^*) = 1$ (thus satisfying the complementary slackness condition (10a)), and

$$E[c(X, Y^*)] = \int_{\{c(x,A)<\frac{1}{\lambda^*}\}} c(x,A)d\mu(x) + \Pr(Z=1)\int_{\left\{c(x,A)=\frac{1}{\lambda^*}\right\}} c(x,A)d\mu(x) + 0 = \underline{c} + p(\bar{c} - \underline{c}) = \delta.$$

This verifies the complementary condition (10b) as well, and hence the coupling described by (X, Y^*) is primal optimal. □

3. Computing Ruin Probabilities: A First Example

In this section, we consider an example with the objective of computing a worst-case estimate of ruin probabilities in a ruin model whose underlying probability measure is not completely specified. As mentioned in the introduction, such a scenario can arise for various reasons, including the lack of data necessary to pin down a satisfactory model. A useful example to keep in mind would be the problem of pricing an exotic insurance contract (e.g., a contract covering extreme climate events) where the probability of ruin has to be maintained below a tolerance level.

Example 1. In this example, we consider the celebrated Cramer–Lundberg model, where a compound Poisson process is used to calculate ruin probabilities of an insurance risk/reserve process. The model is specified by four primitives: initial reserve u, safety loading $\eta > 0$, the rate ν at which claims arrive, and the distribution of claim sizes F with first and second moments m_1 and m_2. Then the stochastic process

$$R(t) = u + (1+\eta)\nu m_1 t - \sum_{i=1}^{N_t} X_i,$$

specifies a model for the reserve available at time t. Here, X_n denotes the size of the nth claim, and the collection $\{X_1, X_2, \ldots\}$ is assumed to be independent samples from the distribution F. Furthermore, $\tilde{p} := (1+\eta)\nu m_1$ is the rate at which a premium is received, and N_t is taken to be a Poisson process with rate λ. Then one of the important problems in risk theory is to calculate the probability

$$\psi(u,T) := \Pr\left\{\inf_{t\in[0,T]} R(t) \le 0\right\}$$

that the insurance firm runs into bankruptcy before a specified duration T.

Despite the simplicity of the model, existing results in the literature do not admit simple methods for the computation of $\psi(u,T)$ (see Asmussen and Albrecher [4], Embrechts et al. [24], Rolski et al. [55], and references therein for a comprehensive collection of results). In addition, if the historical data are not adequately available to choose an appropriate distribution for claim sizes, as is typically the case in an exotic insurance situation, it is not uncommon to use a diffusion approximation

$$R_B(t) := u + (1+\eta)\nu m_1 t - (\nu m_1 t + \sqrt{\nu m_2}\, B(t)) = u + \eta\nu m_1 t - \sqrt{\nu m_2}\, B(t)$$

that depends only on first and second moments of the claim size distribution F. Here, the stochastic process $(B(t) : 0 \le t \le T)$ denotes the standard Brownian motion, and

$$\psi_B(u,T) := \Pr\left\{\sup_{t\in[0,T]} \left(\sqrt{\nu m_2}\, B(t) - \eta\nu m_1 t\right) \ge u\right\},$$

is to serve as a diffusion approximation-based substitute for ruin probabilities $\psi(u,T)$. See the seminal works of Iglehart and Harrison (Iglehart [40] and Harrison [36]) for a justification and some early applications of diffusion approximations in computing insurance ruin. Such diffusion approximations have enabled approximate computations of various path-dependent quantities, which may be otherwise intractable. However, as it is difficult to verify the accuracy of the Brownian approximation of ruin probabilities, in this example, we use the framework developed in Section 2 to compute worst-case estimates of ruin probabilities over all probability measures in a neighborhood around the baseline Brownian motion $B(t)$ driving the ruin model $R_B(t)$.

For this purpose, we identify the Polish space where the stochastic processes of our interest live as the Skorokhod space $S = D([0,T],\mathbb{R})$ equipped with the J_1 topology. In other words, $S = D([0,T],\mathbb{R})$ is simply the space of real-valued right-continuous functions with left limits (rcll) defined on the interval $[0,T]$, equipped with the J_1-metric d_{J_1}. Please refer Lemma B.2 in Appendix B for an expression of d_{J_1} and chapter 3 of Whitt [65] for an excellent exposition on the space $D([0,T],\mathbb{R})$. Next, we take the transportation cost (corresponding to the pth-order Wasserstein distance) as

$$c(x,y) = (d_{J_1}(x,y))^p \quad x,y \in S,$$

for some $p \in [1,\infty)$ and the baseline measure μ as the probability measure induced in the path space by the Brownian motion $B(t)$. In addition, if we let

$$A_u := \left\{ x \in S : \sup_{t\in[0,T]} \left(\sqrt{\nu m_2}\, x(t) - \nu m_1 t\right) \geq u \right\},$$

then the following observations are in order: First, the Brownian approximation for ruin probability $\psi_B(u,T)$ equals $\mu(A_u)$. Next, the set A_u is closed (verified in Lemma B.1 in Appendix B), and hence the function $\mathbf{1}_{A_u}(\cdot)$ is upper semicontinuous. Furthermore, it is verified in Lemma B.2 in Appendix B that

$$c(x,A_u) := \inf\{c(x,y) : y \in A_u\} = \left(u - \sup_{t\in[0,T]} \left(\sqrt{\nu m_2} x(t) - \eta \nu m_1 t\right)\right)^p, \quad \text{for } x \notin A_u. \tag{22}$$

As $h(s) = E_\mu[c(X,A_u); c(X,A_u) \leq s]$ is continuous, for any given $\delta > 0$, it follows from Theorem 2(b) that

$$\psi_{rob}(u,T) := \sup\{P(A_u) : d_c(P,\mu) \leq \delta\} = \mu\left\{ x \in S : c(x,A_u) \leq \frac{1}{\lambda^*} \right\}, \tag{23}$$

where the level $1/\lambda^*$ is identified as $h^{-1}(\delta) = \inf\{s \geq 0 : h(s) \geq \delta\}$. As $c(x,A_u)$ admits a simple form as in (22), following (23), the problem of computing $\psi_{rob}(u,T)$ becomes as elementary as

$$\psi_{rob}(u,T) = \Pr\left\{ \sup_{t\in[0,T]} \left(\sqrt{\nu m_2}\, B_t - \eta \nu m_1 t\right) > u - \left(\frac{1}{\lambda^*}\right)^{1/p} \right\} = \psi_B(\tilde{u},T), \tag{24}$$

with $\tilde{u} := u - (\lambda^*)^{-1/p}$. Thus, the computation of worst-case ruin probability, in this instance, corresponds to simply evaluating the probability that a Brownian motion with negative drift crosses a positive level, $\tilde{u}$, smaller than the original level u. In other words, the presence of model ambiguity has manifested itself only in reducing the initial capital to a new level $\tilde{u}$. In addition to the tractability, such interpretation of worst-case ruin probability in terms of first passage to a modified ruin set is an attractive feature of using Wasserstein distances to model distributional ambiguities.

Numerical Illustration. To make the discussion concrete, we consider a numerical example where $T = 100, \nu = 1$, $p = 2$, and the safety loading $\eta = 0.1$. The claim sizes are taken from a distribution F that is not known to the entire estimation procedure. Our objective is to compute the Brownian approximation to the ruin probability $\psi_B(u,T) = \mu(A_u)$, and its worst-case upper bound $\psi_{rob}(u,T) := \sup\{P(A_u) : d_c(\mu,P) \leq \delta\}$. The estimation methodology is data driven in the following sense: we derive the estimates of moments m_1, m_2 and δ from the observed realizations $\{X_1,\ldots,X_N\}$ of claim sizes. While obtaining the estimates for moments m_1 and m_2 is straightforward, to compute δ, we use the claim size samples $\{X_1,\ldots,X_N\}$ to embed realizations of compound Poisson risk process in a Brownian motion, as in Khoshnevisan [44]. Specific details of the algorithm that estimates δ can be found in Appendix A. The estimate of δ obtained is such that the compensated compound Poisson process of interest that models risk in our setup lies within the δ-neighborhood $\{P : d_c(\mu,P) \leq \delta\}$ with

Table 1. Comparison of the ruin probability estimate $\psi_B(u,T)$ and the worst-case upper bound $\psi_{rob}(u,T)$: Example 1. The denominator $\psi_{LD}(u,T)$ is such that $\psi_{LD}(u,t)/\psi(u,T)\to 1,$ as $u\to\infty$.

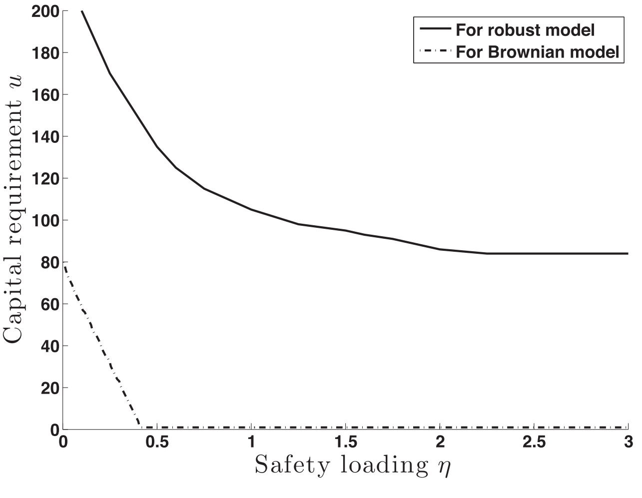
u	$\psi_B(u,T)$	$\tilde{u}$	$\psi_{rob}(u,T)$	$\frac{\psi_B(u,T)}{\psi_{LD}(u,T)}$	$\frac{\psi_{rob}(u,T)}{\psi_{LD}(u,T)}$
50	5.18×10^{-2}	28.23	0.2294	3.32	14.71
100	4.26×10^{-4}	50.78	4.88×10^{-2}	1.07×10^{-1}	12.28
150	4.36×10^{-7}	62.76	1.84×10^{-2}	2.52×10^{-4}	10.65
200	5.05×10^{-11}	69.40	1.02×10^{-2}	5.35×10^{-8}	10.80
250	6.75×10^{-16}	74.29	6.50×10^{-3}	1.15×10^{-12}	10.98

high probability. Next, given the knowledge of Lagrange multiplier λ^*, the evaluations of $\psi_B(u,T)$ and $\psi_{rob}(u,T)=\psi_B(\tilde{u},T)$ (with $\tilde{u}=u-1/\lambda^*$) are straightforward, because of the closed-form expressions for level-crossing probabilities of Brownian motion. Computation of λ^* is also elementary and can be accomplished in multiple ways. We resort to an elementary sample average approximation scheme that solves for λ^*. The estimates of Brownian approximation to ruin probability $\psi_B(u,T)$ and the worst-case ruin probability $\psi_{rob}(u,T)$, for various values of u, are plotted in Table 1. To facilitate comparison, we have used a large deviation approximation of the ruin probabilities $\psi_{LD}(u,T)$ that satisfy $\psi_{LD}(u,T)\sim\psi(u,T)$, as $u\to\infty$, as a common denominator to compare magnitudes of the Brownian estimate $\psi_B(u,T)$ and the worst-case bound $\psi_{rob}(u,T)$. In particular, we have drawn samples of claim sizes from the Pareto distribution $1-F(x)=1\wedge x^{-2.2}$. While the Brownian approximation $\psi_B(u,T)$ remarkably underestimates the ruin probability $\psi(u,T)$, the worst-case upper bound $\psi_{rob}(u,T)$ appears to yield estimates that are conservative yet of the correct order of magnitude.

For further discussion on this toy example, let us say that an insurer paying for the claims distributed according to X_i has a modest objective of keeping the probability of ruin before time T at a level below 0.01. The various combinations of initial capital u and safety loading factors η that achieve this objective are shown in Figure 1. While the (η,u) combinations that work for the Brownian approximation model are depicted by dotted-dashed lines, the corresponding (η,u) combinations that keep $\psi_{rob}(u,T)\le 0.01$ are shown as solid lines.

A Discussion on Regulatory Capital Requirement. For the safety loading $\eta=0.1$ we have considered, the Brownian approximation model $R_B(t)$ requires that the initial capital u be at least 60 to achieve $\psi_B(u,T)\le 0.01$. On the other hand, keeping the robust ruin probability estimate $\psi_{rob}(u,T)$ below 0.01 requires that $u\approx 200$, roughly three times more than the capital requirement of the Brownian model. From the insurer's point of view, there is model uncertainty inherent in contract work, so it is not uncommon to increase the premium income or initial capital by a factor of 3 or 4 (referred to as the "hysteria factor"; see Embrechts et al. [26] and Lewis [46]). This increase in the premium can be thought of as a guess for the price for statistical uncertainty. Choosing a higher premium and capital (larger η and u) along the curve drawn as a solid line in Figure 1, as dictated by the robust estimates of ruin probabilities $\psi_{rob}(u,T)$, instead provides a mathematically sound way

Figure 1. Safety loading versus the capital requirement for the Brownian model $R_B(t)$ (dashed-dotted line) and its robust counterpart (solid line) in Example 1. The objective is to keep the probability of ruin below 0.01.



of doing the same. Unlike the Brownian approximation model, the robust ruin model capital requirement demonstrates that one cannot decrease the probability of ruin by arbitrarily increasing the premium alone; the initial capital also has to be sufficiently large. This prescription is consistent with the behavior of markets with heavy-tailed claims where, in the absence of initial capital, a few initial large claims that occur before enough premium income gets accrued are enough to cause ruin. The minimum capital requirement u prescribed according to the worst-case estimates $\psi_{rob}(u,T)$ can be thought of as the regulatory minimum capital requirement.

Remark 4. Defining the candidate set of ambiguous probability measures via KL divergence or other likelihood-based divergence has been the most common approach while studying distributional robustness (see, e.g., Hansen and Sargent [35], Ben-Tal et al. [11], and Breuer and Csiszár [18]). This would not have been useful in this setup, as the compensated Poisson process that models risk is not absolutely continuous with respect to Brownian motion. In general, it is normal to run into absolute continuity issues when we deal with continuous-time stochastic processes. In such instances, as demonstrated in Example 1, modeling ambiguity via optimal transport costs or Wasserstein distances not only offers a tractable alternative but also yields insightful equivalent reformulations.

4. Proof of the Duality Theorem

4.1. Duality in Compact Spaces

We prove Theorem 1 by first proving the following progressively strong duality results in Polish spaces S that are compact. Proofs of all the technicalities that are not central to the argument are relegated to Appendix B.

Proposition 1. *Suppose that S is a compact Polish space, and $f:S\to\mathbb{R}$ satisfies Assumption (A2). In addition to satisfying Assumption (A1), suppose that $c:S\times S\to\mathbb{R}_+$ is continuous. Then $\bar{I}=\underline{J}$, and a primal optimizer $\pi^*\in\Phi_{\mu,\delta}$ satisfying $I(\pi^*)=\bar{I}$ exists.*

Proposition 2. *Suppose that S is a compact Polish space, $f:S\to\mathbb{R}$ satisfies Assumption (A2), and $c:S\times S\to[0,+\infty]$ satisfies Assumption (A1). Then $\bar{I}=\underline{J}$, and a primal optimizer $\pi^*\in\Phi_{\mu,\delta}$ satisfying $I(\pi^*)=\bar{I}$ exists.*

As in the proof of the standard Kantorovich duality in compact spaces in Villani [61], we use Fenchel duality theorem (see theorem 1 in chapter 7 of Luenberger [48]) to prove Proposition 1. However, the similarity stops there, owing to the reason that the set of primal feasible measures $\Phi_{\mu,\delta}$ is not tight when S is noncompact (contrast this with Kantorovich duality, where the collection of feasible measures $\Pi(\mu,\nu)$, the set of all joint distributions between any two probability measures μ, and ν is tight).

Proof of Proposition 1. As a preparation for applying Fenchel duality theorem, we first let $X=C_b(S\times S)$, and we identify $X^*=M(S\times S)$ as its topological dual. Here, the spaces X and X^*, representing the vector space of bounded continuous functions and finite Borel measures on $S\times S$, respectively, are equipped with the supremum and total variation norms. The fact that X^* is the dual space of X is a consequence of the Riesz representation theorem (see, e.g., Rudin [56]). Next, let C be the collection of all functions g in X that are of the form

$$g(x,y)=\varphi(x)+\lambda c(x,y),\qquad \text{for all } x,y,$$

for some $\varphi\in C_b(S)$ and $\lambda\geq 0$. In addition, let D denote the collection of all functions g in X that satisfy $g(x,y)\geq f(y)$ for every x,y. Every g in the convex subset C is defined by the pair (λ,φ), which, in turn, can be uniquely identified by

$$\varphi(x)=g(x,x)\quad\text{and}\quad \lambda=\frac{g(x,y)-\varphi(x)}{c(x,y)},$$

for some x,y in S such that $c(x,y)\neq 0$. Keeping this invertible relationship in mind, define the functionals $\Phi:C\to\mathbb{R}$ and $\Gamma:D\to\mathbb{R}$ as below:

$$\Phi(g):=\lambda\delta+\int\varphi d\mu\quad\text{and}\quad\Gamma(g):=0.$$

Evidently, the functional Φ is convex, Γ is concave, and we are interested in

$$\inf_{g\in C\cap D}(\Phi(g)-\Gamma(g))=\inf\{J(\lambda,\varphi):\lambda\geq 0,\ \varphi\in C_b(S),\ \varphi(x)+\lambda c(x,y)\geq f(y)\ \text{for all } x,y\}. \tag{25}$$

The next task is to identify the conjugate functionals Φ^*, Γ^* and their respective domains C^*, D^* as below:

$$C^* = \left\{\pi \in X^* : \sup_{g\in C}\left\{\int g d\pi - \Phi(g)\right\} < \infty\right\} \quad \text{and} \quad D^* = \left\{\pi \in X^* : \inf_{g\in D}\int g d\pi > -\infty\right\}.$$

The conjugate functionals $\Phi^* : C^* \to \mathbb{R}$ and $\Gamma^* : D^* \to \mathbb{R}$ are defined accordingly as

$$\Phi^*(\pi) := \sup_{g\in C}\left\{\int g d\pi - \Phi(g)\right\} \quad \text{and} \quad \Gamma^*(\pi) := \inf_{g\in D}\int g d\pi.$$

First, to determine C^*, we see that for every $\pi \in M(S\times S)$,

$$\begin{aligned}\sup_{g\in C}\left\{\int g d\pi - \Phi(g)\right\} &= \sup_{(\lambda,\varphi)\in\mathbb{R}_+\times C_b(S)}\left\{\int (\varphi(x) + \lambda c(x,y))d\pi(x,y) - \left(\lambda\delta + \int \varphi(x)d\mu(x)\right)\right\}\\ &= \sup_{(\lambda,\varphi)\in\mathbb{R}_+\times C_b(S)}\left\{\lambda\left(\int c(x,y)d\pi(x,y) - \delta\right) + \int \varphi(x)(d\pi(x,y) - d\mu(x))\right\},\\ &= \begin{cases} 0 & \text{if } \int c d\pi \le \delta \text{ and } \pi(A\times S) = \mu(A) \text{ for all } A\in\mathscr{B}(S),\\ \infty & \text{otherwise.}\end{cases}\end{aligned}$$

Therefore,

$$C^* = \left\{\pi \in M(S\times S) : \int c d\pi \le \delta,\ \pi(A\times S) = \mu(A) \ \text{ for all } \ A\in\mathscr{B}(S)\right\} \quad \text{and} \quad \Phi^*(\pi) = 0.$$

Next, to identify D^*, we show in Lemma B.6 in Appendix B that $\inf_{g\in D}\int g d\pi = -\infty$ if a measure $\pi\in M(S\times S)$ is not nonnegative. On the other hand, if $\pi \in M(S\times S)$ is nonnegative, then

$$\inf\left\{\int g(x,y)d\pi(x,y) : g(x,y) \ge f(y) \text{ for all } x,y\right\} = \int f(y)d\pi(x,y).$$

This is because f, being an upper semicontinuous function that is bounded from above, can be approximated pointwise by a monotonically decreasing sequence of continuous functions.[6] Then the above equality follows as a consequence of monotone convergence theorem. As a result,

$$D^* = \left\{\pi \in M_+(S\times S) : \int f(y)d\pi(x,y) > -\infty\right\} \quad \text{and} \quad \Gamma^*(\pi) = \int f(y)d\pi(x,y).$$

Then, $\Gamma^*(\pi) - \Phi^*(\pi) = \int f(y)d\pi(x,y)$ on $C^*\cap D^* = \{\pi \in \cup_{\nu\in P(S)}\Pi(\mu,\nu) : \int c d\pi \le \delta, \int f(y)d\pi(x,y) > -\infty\}$. As $\bar{I}$ is defined to equal $\sup\{\int f d\nu : d_c(\mu,\nu) \le \delta, \int f d\nu > -\infty\}$, it follows that

$$\sup\{\Gamma^*(\pi) - \Phi^*(\pi) : \pi \in C^*\cap D^*\} = \bar{I}. \tag{26}$$

As the set $C\cap D$ contains points in the relative interiors of C and D (consider, e.g., the candidate $h(x,y) = c(x,y) + \sup_{x\in S} f(x)$, where $\sup_{x\in S} f(x) < \infty$ because f is upper semicontinuous and S is compact), and the epigraph of the function Γ has nonempty interior, it follows as a consequence of Fenchel's duality theorem (see, e.g., theorem 1 in chapter 7 of Luenberger [48]) that

$$\inf_{g\in C\cap D}(\Phi(g) - \Gamma(g)) = \sup\{\Gamma^*(\pi) - \Phi^*(\pi) : \pi \in C^*\cap D^*\},$$

where the supremum in the right is achieved by some $\pi^* \in \Phi_{\mu,\delta}$. In other words (see (25) and (26)),

$$\inf\{J(\lambda,\varphi) : \lambda \ge 0, \varphi\in C_b(S),\ \varphi(x) + \lambda c(x,y) \ge f(y) \text{ for all } x,y\} = \max\{I(\pi) : \pi\in\Phi_{\mu,\delta}\} =: \bar{I}.$$

As $C_b(S) \subseteq \mathfrak{m}_{\mathcal{U}}(S; \mathbb{R}\cup\{+\infty\})$, it follows from the definition of $\underline{J}$ that

$$\underline{J} \le \inf\{J(\lambda,\varphi) : \lambda \ge 0,\ \varphi\in C_b(S),\ \varphi(x) + \lambda c(x,y) \ge f(y) \ \text{ for all } x,y\} = \bar{I}.$$

However, because of weak duality (5), we have $\underline{J} \ge \bar{I}$. Therefore, $\underline{J} = \bar{I} = \max\{I(\pi) : \pi\in\Phi_{\mu,\delta}\}$. □

Proof of Proposition 2. First, observe that c is a nonnegative lower semicontinuous function defined on the Polish space $S\times S$. Therefore, we can write $c=\sup_n c_n$, where $(c_n:n\geq 1)$ is a nondecreasing sequence of continuous cost functions $c_n:S\times S\to\mathbb{R}_+$.[7] With c_n as the cost function, define the corresponding primal and dual problems:

$$\bar{I}_n=\sup_{\pi\in\Phi^n_{\mu,\delta}} I(\pi)\quad\text{and}\quad \underline{J}_n=\inf_{(\lambda,\varphi)\in\Lambda_{c_n,f}} J(\lambda,\varphi).$$

Here, the feasible sets $\Phi^n_{\mu,\delta}$ and $\Lambda^n_{c,f}$ are obtained by suitably modifying the cost function in sets $\Phi_{\mu,\delta}$ and $\Lambda_{c,f}$ (defined in (6a) and (6b)) as below:

$$\Phi^n_{\mu,\delta}:=\left\{\pi\in\bigcup_{\nu\in P(S)}\Pi(\mu,\nu):\int c_n d\pi\leq\delta\right\}\quad\text{and}$$

$$\Lambda_{c_n,f}:=\{(\lambda,\varphi):\lambda\geq 0,\varphi\in\mathfrak{m}_{\mathcal{U}}(S;\mathbb{R}\cup\{+\infty\}),\varphi(x)+\lambda c_n(x,y)\geq f(y)\text{ for all }x,y\in S\}.$$

As the cost functions c_n are continuous, because of Proposition 1, there exists a sequence of measures $(\pi_n:n\geq 1)$ such that

$$I(\pi_n)=\bar{I}_n=\underline{J}_n. \tag{27}$$

As S is compact, the set $(\pi_n:n\geq 1)$ is automatically tight, and because of Prokhorov's theorem, there exists a subsequence $(\pi_{n_k}:k\geq 1)$ weakly converging to some $\pi^*\in P(S\times S)$. First, we check that π^* is feasible:

$$\int c d\pi^*=\int\lim_n c_n d\pi^*=\lim_n\int c_n d\pi^*=\lim_n\lim_k\int c_n d\pi_{n_k},$$

where the second and third equalities are consequences of monotone convergence and weak convergence, respectively. In addition, since $(c_n:n\geq 1)$ is a nondecreasing sequence of functions,

$$\lim_k\int c_n d\pi_{n_k}\leq\underline{\lim}_k\int c_{n_k}d\pi_{n_k}\leq\delta.$$

Therefore, $\int c d\pi^*\leq\delta$. Again, as a simple consequence of weak convergence, we have $\int g(x)d\pi^*(x,y)=\lim_n\int g(x)d\pi_{n_k}(x,y)=\int g(x)d\mu(x)$, for every $g\in C_b(S)$. Therefore, $\pi^*\in\Phi_{\mu,\delta}$, and is indeed feasible. Next, as f is upper semicontinuous and S is compact, f is bounded from above. Then, because of the weak convergence $\pi_{n_k}\Rightarrow\pi^*$, the objective function $I(\pi^*)$ satisfies

$$I(\pi^*)=\int f\,d\pi^*\geq\overline{\lim_k}\int f\,d\pi_{n_k}=\overline{\lim_k}\,I(\pi_{n_k})=\overline{\lim_k}\,\underline{J}_{n_k},$$

because $I(\pi_{n_k})=\underline{J}_{n_k}$ as in (27). Furthermore, as $c_n\leq c$, it follows that $\Lambda_{c_n,f}$ is a subset of $\Lambda_{c,f}$, and hence $\underline{J}_n\geq\underline{J}$, for all n. As a result, we obtain

$$I(\pi^*)\geq\overline{\lim_k}\,\underline{J}_{n_k}\geq\underline{J}.$$

Since $\bar{I}$ never exceeds $\underline{J}$ (see (5)), it follows that $I(\pi^*)=\bar{I}=\underline{J}$. □

4.2. A Note on Additional Notations and Measurability

Given that we have established duality in compact spaces, the next step is to establish the same when S is not compact. For this purpose, we need the following additional notation. For any probability measure $\pi\in P(S\times S)$, let

$$S_\pi:=\mathrm{Spt}(\pi_X)\cup\mathrm{Spt}(\pi_Y),$$

where $\mathrm{Spt}(\pi_X)$ and $\mathrm{Spt}(\pi_Y)$ denote the respective supports of marginals $\pi_X(\cdot):=\pi(\cdot\times S)$ and $\pi_Y(\cdot):=\pi(S\times\cdot)$. As every probability measure defined on a Polish space has σ-compact support, the set $S_\pi\times S_\pi$, which is a subset of $S\times S$, is σ-compact. As we shall see in the proof of Proposition 3, $S_\pi\times S_\pi$ can be written as the union of an increasing sequence of compact subsets $(S_n\times S_n:n\geq 1)$. It will then be easy to make progress toward strong duality in noncompact sets such as $S_\pi\times S_\pi$ by utilizing the strong duality results derived in Section 4.1 (which are applicable for compact sets $S_n \times S_n$) via a sequential argument; this is accomplished in Section 4.3 after introducing additional notation as follows. Recall our earlier definition that $C := \{(x,y) \in S \times S : c(x,y) < \infty\}$. For any closed $K \subseteq S$, let

$$\Lambda(K \times K) := \{(\lambda, \varphi) : \lambda \geq 0, \varphi \in \mathfrak{m}_{\mathcal{U}}(K; \mathbb{R} \cup \{+\infty\}),\ \varphi(x) + \lambda c(x,y) \geq f(y) \text{ for all } (x,y) \in (K \times K) \cap C\}, \tag{28}$$

where $\mathfrak{m}_{\mathcal{U}}(K; \mathbb{R} \cup \{+\infty\})$ is used to denote the collection of measurable functions $\varphi : (K, \mathcal{U}(K)) \to (\mathbb{R} \cup \{+\infty\}, \mathcal{B}(\mathbb{R} \cup \{+\infty\}))$, with $\mathcal{U}(K)$ denoting the universal σ–algebra of the Polish space K. With this notation, the dual feasible set $\Lambda_{c,f}$ is nothing but $\Lambda(S \times S)$.

The next issue we address is the measurability of functions of the form $\varphi_\lambda(x) = \sup_{y \in S}\{f(y) - \lambda c(x,y)\}$. Here, recall our notational convention that $\lambda c(x,y) = +\infty$ if $\lambda = 0$ and $c(x,y) = +\infty$. For any $u \in \mathbb{R}$ and $\lambda \in [0,\infty)$, $\{\varphi_\lambda(x) > u\} = \mathrm{Proj}_1(\{(x,y) \in C : f(y) - \lambda c(x,y) > u\})$, which is analytic, because projections $\mathrm{Proj}_i(A) := \{x_i \in S : (x_1, x_2) \in A\}, i = 1, 2$ of any Borel-measurable set A are analytic (see, e.g., proposition 7.39 in Bertsekas and Shreve [13]). As analytic subsets of S lie in $\mathcal{U}(S)$ (see corollary 7.42.1 in Bertsekas and Shreve [13]), the function $\varphi_\lambda(x) : (S, \mathcal{U}(S)) \to (\overline{\mathbb{R}}, \mathcal{B}(\overline{\mathbb{R}}))$ is measurable (i.e., universally measurable; refer chapter 7 of Bertsekas and Shreve [13] for an introduction to universal measurability and analytic sets).

Finally, let E denote the convex set of probability measures defined as follows:

$$E := \left\{\pi \in \bigcup_{\nu \in P(S)} \Pi(\mu, \nu) : \int c(x,y) d\pi(x,y) < \infty,\ \int f(y) d\pi(x,y) \in (-\infty, \infty)\right\}.$$

4.3. Extension of Duality to Noncompact Spaces

Proposition 3 is the first step toward extending the strong duality results proved in Section 4.1 to noncompact sets.

Proposition 3. *Suppose that Assumptions (A1) and (A2) are in force. Then, for any $\pi \in E$,*

$$\inf_{(\lambda,\varphi) \in \Lambda(S_\pi \times S_\pi)} J(\lambda, \varphi) \leq \bar{I}.$$

Proof of Proposition 3. As any Borel probability measure on a Polish space is concentrated on a σ-compact set (see theorem 1.3 of Billingsley [15]), the sets $\mathrm{Spt}(\pi_X)$ and $\mathrm{Spt}(\pi_Y)$ are σ-compact, and therefore $S_\pi \times S_\pi$ is σ-compact as well. Then, by the definition of σ-compactness, the set $S_\pi \times S_\pi$ can be written as the union of an increasing sequence of compact subsets $(C_n : n \geq 1)$ of $S_\pi \times S_\pi$. If we let S_n to be the union of the projections of C_n over its two coordinates, then it follows that $(S_n \times S_n : n \geq 1)$ is an increasing sequence of compact subsets of $S_\pi \times S_\pi$ satisfying $\mathrm{Spt}(\pi) \subseteq S_\pi \times S_\pi = \cup_{n \geq 1}(S_n \times S_n)$. As $\int |f(y)| d\pi(x,y)$ and $\int c(x,y) d\pi(x,y)$ are finite, one can pick the increasing sequence $(S_n \times S_n : n \geq 1)$ with $S_\pi \times S_\pi = \cup_{n \geq 1}(S_n \times S_n)$ to be satisfying,

$$p_n := \pi(S_n \times S_n) \geq 1 - \frac{1}{n}, \tag{29a}$$

$$\int c(x,y) \mathbf{1}_{(S_n \times S_n)^c} d\pi(x,y) \leq \frac{\delta}{n}, \quad \text{and} \tag{29b}$$

$$\int |f|(y) \mathbf{1}_{(S_n \times S_n)^c} d\pi(x,y) \leq \frac{1}{n}, \tag{29c}$$

where $(S_n \times S_n)^c := (S \times S) \backslash (S_n \times S_n)$, and $\delta \in (0, \infty)$ is introduced while defining the primal feasibility set $\Phi_{\mu,\delta}$ (recall that $\bar{I} := \sup\{I(\pi) : \pi \in \Phi_{\mu,\delta}\}$) and the dual objective $J(\lambda, \varphi) = \lambda\delta + \int \varphi d\mu$ in Section 2.2. Having chosen the compact subsets $(S_n : n \geq 1)$, define a joint probability measure $\pi_n \in P(S_n \times S_n)$ and its corresponding marginal $\mu_n \in P(S_n)$ as below:

$$\pi_n(\cdot) = \frac{\pi(\cdot \cap (S_n \times S_n))}{p_n} \quad \text{and} \quad \mu_n(\cdot) = \pi_n(\cdot \times S_n),$$

for every $n \geq 1$. Additionally, let $\delta_n := \delta(1-1/n)$. For every $n \geq 1$, these new definitions can be used to define "restricted" primal and dual optimization problems $\bar{I}_n$ and $\underline{J}_n$ supported on the compact set $S_n \times S_n$:

$$\bar{I}_n := \sup\left\{\int f(y)\gamma(dx,dy) : \gamma \in \bigcup_{\nu \in P(S_n)} \Pi(\mu_n, \nu), \int c(x,y)d\gamma(x,y) \leq \delta_n\right\} \text{ and } \tag{30}$$

$$\underline{J}_n := \inf\left\{\lambda\delta_n + \int \varphi d\mu_n : (\lambda, \varphi) \in \Lambda(S_n \times S_n)\right\}. \tag{31}$$

To comprehend (31), refer to the definition of $\Lambda(\cdot)$ in (28). As S_n is compact, because of the duality result in Proposition 2, we know that there exists a $\gamma_n^* \in P(S_n \times S_n)$ that is feasible for the optimization of $\bar{I}_n$ and satisfies

$$\int f(y)\gamma_n^*(dx,dy) = \bar{I}_n = \underline{J}_n. \tag{32}$$

Using this optimal measure γ_n^*, one can, in turn, construct a measure $\tilde{\pi} \in P(S \times S)$ by stitching it together with the residual portion of π as below:

$$\tilde{\pi}(\cdot) = p_n\gamma_n^*(\cdot \cap (S_n \times S_n)) + \pi(\cdot \cap (S_n \times S_n)^c).$$

Having tailored a candidate measure $\tilde{\pi} \in P(S \times S)$, we next check its feasibility that $\tilde{\pi} \in \Phi_{\mu,\delta}$ as follows: since $\gamma_n^* \in \Pi(\mu_n, \nu)$ for some $\nu \in P(S_n)$, it is easy to check, as below, that $\tilde{\pi}$ has μ as its marginal for the first component. It follows from the definition of $\tilde{\pi}$ that

$$\tilde{\pi}(\cdot \times S) = p_n\gamma_n^*((\cdot \cap S_n) \times S_n) + \pi((\cdot \times S) \cap (S_n \times S_n)^c) = p_n\mu_n(\cdot \cap S_n) + \pi((\cdot \times S) \cap (S_n \times S_n)^c).$$

As $\mu_n(\cdot) = \pi(\cdot \times S_n)/p_n$, it then follows that $p_n\mu_n(\cdot \cap S_n) = \pi((\cdot \times S) \cap (S_n \times S_n))$, and consequently,

$$\tilde{\pi}(\cdot \times S) = \pi((\cdot \times S) \cap (S_n \times S_n)) + \pi((\cdot \times S) \cap (S_n \times S_n)^c) = \pi(\cdot \times S) = \mu(\cdot).$$

Therefore, $\tilde{\pi} \in \Pi(\mu, \nu)$ for some $\nu \in P(S)$. Furthermore, as γ_n^* is a feasible solution for the optimization problem in (30),

$$\int c d\tilde{\pi} = p_n \int_{S_n \times S_n} c d\gamma_n^* + \int_{(S_n \times S_n)^c} c d\pi \leq \delta_n + \frac{\delta}{n},$$

which does not exceed δ. To see this, refer to (29b), and recall that $\delta_n := \delta(1-1/n)$. Therefore, the stitched measure $\tilde{\pi}$ is primal feasible; that is, $\tilde{\pi} \in \Phi_{\mu,\delta}$. Consequently,

$$\bar{I} \geq I(\tilde{\pi}) = \int f(y)d\tilde{\pi}(x,y) = p_n \int f(y)d\gamma_n^*(x,y) + \int f(y)\mathbf{1}_{(S_n \times S_n)^c}\, d\pi(x,y).$$

As S_n is chosen to satisfy (29c), we have $\bar{I} \geq p_n\bar{I}_n - n^{-1}$. Therefore, it is immediate from (32) that

$$\underline{J}_n \leq \left(1 - \frac{1}{n}\right)^{-1}\left(\bar{I} + \frac{1}{n}\right), \tag{33}$$

which, if $\bar{I} < \infty$, is a useful upper bound for all of $\underline{J}_n, n \geq 1$. See that the statement of Proposition 3 already holds when $\bar{I}$ equals ∞. Therefore, let us take $\bar{I}$ to be finite. Next, given $n \geq 1$ and $\varepsilon > 0$, take an ε-optimal solution (λ_n, φ_n) for $\underline{J}_n$: that is, $(\lambda_n, \varphi_n) \in \Lambda(S_n \times S_n)$ and

$$\lambda_n\delta_n + \int \varphi_n d\mu_n \leq \underline{J}_n + \varepsilon.$$

For any pair (λ_n, φ) that belongs to $\Lambda(S_n \times S_n)$, we have from the definition of $\Lambda(S_n \times S_n)$ that $\varphi(x)$ is larger than $\sup_{z \in S_n}\{f(z) - \lambda_n c(x,z)\}$, for every $x \in S_n$:

$$\lambda_n\delta_n + \int \sup_{z \in S_n}\{f(z) - \lambda_n c(x,z)\}d\mu_n(x) \leq \underline{J}_n + \varepsilon,$$

for every n. Here, recall our notational convention that $\lambda_n c(x,y) = +\infty$ if $\lambda_n = 0$ and $c(x,y) = +\infty$. As $\mu_n(\cdot) = \pi(\cdot \times S_n)/p_n$, combining the above expression with (33) results in

$$\overline{\lim_{n\to\infty}} \left(\lambda_n \delta_n + \int \sup_{z \in S_n} \{ f(z) - \lambda_n c(x,z) \} \frac{\mathbf{1}_{S_n}(x)\mathbf{1}_{S_n}(y)}{p_n} d\pi(x,y) \right) \le \bar{I} + \varepsilon. \tag{34}$$

Since $c(x,x) = 0$, the integrand admits $f(x)$ as a lower bound on $S_n \times S_n$; furthermore, as $f \in L^1(d\mu)$, $-f^-(x)$ serves as a common integrable lower bound for all $n \ge 1$. This has two consequences: The first is that, because of the finite upper bound in (34), $\underline{\lim}_n \lambda_n$ and $\overline{\lim}_n \lambda_n$ are finite (recall that $\lambda_n \ge 0$ as well), and there exists a subsequence $(\lambda_{n_k} : k \ge 1)$ such that $\lambda_{n_k} \to \lambda^*$, as $k \to \infty$, for some $\lambda^* \in [0, \infty)$. The second consequence of the existence of a common integrable lower bound is that we can apply Fatou's lemma and dominated convergence in (34) along the subsequence $(n_k : k \ge 1)$ to obtain

$$\begin{aligned} \bar{I} + \varepsilon &\ge \underline{\lim}_{k\to\infty} \left(\lambda_{n_k} \delta_{n_k} + \int \sup_{z \in S_{n_k}} \{ f(z) - \lambda_{n_k} c(x,z) \} \frac{\mathbf{1}_{S_{n_k}}(x)\mathbf{1}_{S_{n_k}}(y)}{p_{n_k}} d\pi(x,y) \right) \\ &\ge \lambda^* \delta + \int_{S_\pi \times S_\pi} \sup_{z \in \cup_k S_{n_k}} \{ f(z) - \lambda^* c(x,z) \} d\pi(x,y) \\ &= \lambda^* \delta + \int_S \sup_{z \in S_\pi} \{ f(z) - \lambda^* c(x,z) \} d\mu(x). \end{aligned}$$

Here, we have used that $\delta_n \to \delta, p_n \to 1$ as $n \to \infty$, $\cup_{n \ge 1} S_n = \cup_{k \ge 1} S_{n_k} = S_\pi$, and the fact that π is supported on a subset of $S_\pi \times S_\pi$. The fact that $\underline{\lim}_k \sup_{z \in S_{n_k}} \{ f(z) - \lambda_{n_k} c(x,z) \}$ is at least $\sup_{z \in \cup_k S_{n_k}} \{ f(z) - \lambda^* c(x,z) \}$, for every $x \in S$, is carefully checked in Lemma B.7 in Appendix B. If we let $\varphi^*(x) := \sup_{y \in S_\pi} \{ f(y) - \lambda^* c(x,y) \}$, then $(\lambda^*, \varphi^*) \in \Lambda(S_\pi \times S_\pi)$, and as $\varepsilon > 0$ is arbitrary, it follows from the above inequality that $J(\lambda^*, \varphi^*) \le \bar{I}$. This completes the proof. □

Proposition 4. *Suppose that Assumptions* (A1) *and* (A2) *are in force. Then for any* $\lambda \ge 0$,

$$\sup_{\pi \in E} \int (f(y) - \lambda c(x,y)) d\pi(x,y) = \int \sup_{y \in S} \{ f(y) - \lambda c(x,y) \} d\mu(x).$$

Proof of Proposition 4. Recalling our notational convention that $\lambda c(x,y) = +\infty$ if $\lambda = 0$ and $c(x,y) = +\infty$, let $g(x,y) := f(y) - \lambda c(x,y)$. For $n \ge 1$ and $k \le n^2$, define $A_{k,n} := \{ (x,y) \in S \times S : (k-1)/n \le g(x,y) \le k/n \}$ and $B_{k,n} := \mathrm{Proj}_1(A_{k,n}) \backslash \cup_{j > k} \mathrm{Proj}_1(A_{j,n})$. Here, recall that for any $A \subset S \times S, \mathrm{Proj}_1(A) = \{ x_1 : (x_1, x_2) \in A \}$. The sequence $(B_{k,n} : k \le n^2)$ also admits the following equivalent backward recursive definition:

$$B_{n^2,n} = \mathrm{Proj}_1(A_{n^2,n}), \quad \text{and} \quad B_{k,n} = \mathrm{Proj}_1(A_{k,n}) \backslash \cup_{j>k} B_{j,n} \ \text{ for } k < n^2,$$

that renders the collection $(B_{k,n} : k \le n^2)$ disjoint. As the function g is upper semicontinuous, it is immediate that the sets $A_{k,n}$ are Borel-measurable and their corresponding projections $B_{k,n}$ are analytic subsets of S. Then, because of the Jankov–von Neumann selection theorem,[8] there exists, for each $k \le n^2$ such that $A_{k,n} \neq \varnothing$, a universally measurable function $\gamma_k(x) : \mathrm{Proj}_1(A_{k,n}) \to S$ such that $(x, \gamma_k(x)) \in A_{k,n}$, or in other words,

$$\frac{k-1}{n} \le g(x, \gamma_k(x)) \le \frac{k}{n}. \tag{35}$$

Next, let us define the function $\Gamma_n : S \to S$ as below:

$$\Gamma_n(x) = \begin{cases} \gamma_k(x), & \text{if } x \in B_{k,n} \text{ for some } k \le n^2 \\ x, & \text{otherwise.} \end{cases}$$

This definition is possible because $B_{k,n} \subseteq \mathrm{Proj}_1(A_{k,n})$, and the collection $(B_{k,n} : k \le n^2)$ is disjoint. As each γ_k is universally measurable, the composite function $\Gamma_n(x)$ is universally measurable as well and satisfies that $\sup \{ g(x,y) : g(x,y) \le n, y \in S \} - 1/n \le g(x, \Gamma_n(x)) \le f(x) \vee n$ for each x. Both sides of the inequality above can be inferred from (35) after recalling that $g(x,x) = f(x) - \lambda c(x,x) = f(x)$. Letting $n \to \infty$, we see that

$$\sup \{ g(x,y) : y \in S \} \le \underline{\lim}_n g(x, \Gamma_n(x)) \le \infty. \tag{36}$$

Next, define the family of probability measures $(\pi_n : n \geq 1)$ as $d\pi_n(x,y) := d\mu(x) \times d\delta_{\Gamma_n(x)}(y)$, with $\delta_a(\cdot)$ representing the Dirac measure centered at $a \in S$. As $g(x,y) \leq f(x) \vee n, \pi_n$ almost surely, we have that the functions $c(x,y)$ and $f(y)$ are integrable with respect to π_n. In addition, since $\pi_n(\cdot \times S) = \mu(\cdot)$, we have $\pi_n \in E$. Finally, since $g(x, \Gamma_n(x)) \geq f(x) - 1$ for all n, it follows from Fatou's lemma and (36) that

$$\varliminf_{n\to\infty} \int g(x,y) d\pi_n(x,y) = \varliminf_{n\to\infty} \int g(x,\Gamma_n(x)) d\mu(x) \geq \int \varliminf_{n\to\infty} g(x,\Gamma_n(x)) d\mu(x) \geq \int \sup_{y\in S} g(x,y) d\mu(x),$$

which proves that $\sup_{\pi\in E} \int g d\pi \geq \int \sup_{y\in S} g(x,y) d\mu(x)$. Since $\int g d\pi \leq \int \sup_{y\in S} g(x,y) d\pi(x,y)$ for any $\pi \in E$, the proof of Proposition 4 is complete. □

Proof of Theorem 1(a). If $\bar{I} = \infty$, then as $\bar{I}$ never exceeds $\underline{J}, \underline{J}$ equals ∞ as well, and there is nothing to prove. Next, consider the case where $\bar{I}$ is finite; that is, $\bar{I} \in [\int f d\mu, \infty)$. Then, because of Proposition 3, for every $\pi \in E$,

$$\bar{I} \geq \inf_{(\lambda,\varphi)\in\Lambda(S_\pi\times S_\pi)} \left\{ \lambda\delta + \int \varphi(x) d\mu(x) \right\} \geq \inf_{\lambda\geq 0} \left\{ \lambda\delta + \int \sup_{y\in S_\pi} \{f(y) - \lambda c(x,y)\} d\mu(x) \right\},$$

with the notational convention that $\lambda c(x,y) = +\infty$ if $\lambda = 0$ and $c(x,y) = +\infty$. For any $\pi \in E$ and $\lambda \geq 0$, define

$$T(\lambda,\pi) := \lambda\delta + \int \sup_{y\in S_\pi} \{f(y) - \lambda c(x,y)\} d\mu(x).$$

As $c(x,x) = 0$, it is easy to see that $T(\lambda,\pi) \geq \lambda\delta + \int f\, d\mu$. Since $T(\lambda,\pi) > \bar{I}$ for every $\lambda > \lambda_{\max} := (\bar{I} - \int f\, d\mu)/\delta$, one can rather restrict attention to the compact subset $[0, \lambda_{\max}]$, as follows:

$$\bar{I} \geq \inf_{\lambda\in[0,\lambda_{\max}]} T(\lambda,\pi). \tag{37}$$

Being a pointwise supremum of a family of affine functions, $\sup_{y\in S_\pi}\{f(y) - \lambda c(x,y)\}$ is a lower semicontinuous function with respect to the variable λ. Furthermore, recall that $T(\lambda,\pi) \geq \lambda\delta + \int f\, d\mu$. Thus, for any $\lambda_n \to \lambda$, because of Fatou's lemma,

$$\varliminf_{n\to\infty} T(\lambda_n,\pi) \geq \lambda\delta + \int \varliminf_{n} \sup_{y\in S_\pi}\{f(y) - \lambda_n c(x,y)\} d\mu(x) \geq \lambda\delta + \int \sup_{y\in S_\pi}\{f(y) - \lambda c(x,y)\} d\mu(x) = T(\lambda,\pi),$$

which means that $T(\lambda,\pi)$ is lower semicontinuous in λ. Along with this lower semicontinuity, for every fixed $\pi, T(\lambda,\pi)$ is also convex in λ. In addition, for any $\alpha \in (0,1)$ and $\pi_1, \pi_2 \in E$,

$$\begin{aligned} T(\lambda, \alpha\pi_1 + (1-\alpha)\pi_2) &= \lambda\delta + \int \sup_{y\in S_{\alpha\pi_1+(1-\alpha)\pi_2}} \{f(y) - \lambda c(x,y)\} d\mu(x) \\ &\geq \max_{i=1,2} \left\{ \lambda\delta + \int \sup_{y\in S_{\pi_i}}\{f(y) - \lambda c(x,y)\} d\mu(x) \right\} \geq \alpha T(\lambda,\pi_1) + (1-\alpha) T(\lambda,\pi_2). \end{aligned}$$

Therefore, $T(\lambda,\pi)$ is a concave function in π for every fixed λ. One can apply a standard minimax theorem (see, e.g., Sion [59]) to conclude that

$$\sup_{\pi\in E} \inf_{\lambda\in[0,\lambda_{\max}]} T(\lambda,\pi) = \inf_{\lambda\in[0,\lambda_{\max}]} \sup_{\pi\in E} T(\lambda,\pi).$$

This observation, in conjunction with (37), yields

$$\bar{I} \geq \sup_{\pi\in E} \inf_{\lambda\in[0,\lambda_{\max}]} T(\lambda,\pi) = \inf_{\lambda\in[0,\lambda_{\max}]} \left\{ \lambda\delta + \sup_{\pi\in E} \int \sup_{y\in S_\pi}\{f(y) - \lambda c(x,y)\} d\mu(x) \right\}. \tag{38}$$

Now, since $\int \sup_{y\in S_\pi}\{f(y) - \lambda c(x,y)\} d\mu(x) \geq \int (f(y) - \lambda c(x,y)) d\pi(x,y)$ for any $\pi \in E$, it follows from Proposition 4 that

$$\sup_{\pi\in E} \int \sup_{y\in S_\pi}\{f(y) - \lambda c(x,y)\} d\mu(x) = \int \sup_{y\in S}\{f(y) - \lambda c(x,y)\} d\mu(x) = \int \varphi_\lambda(x) d\mu(x).$$

This observation, in conjunction with (38), establishes that $\bar{I} \geq \inf_{\lambda\geq 0}\{\lambda\delta + \int \varphi_\lambda(x)d\mu(x)\}$, which completes the proof of Theorem 1(a).

Proof of Theorem 1(b). Recall that $\int f\,d\mu$ is finite. If we let $g(\lambda) := \lambda\delta + \int \varphi_\lambda(x)d\mu(x)$, then because of a routine application of Fatou's lemma, we have that $\underline{\lim}_{n\to\infty} g(\lambda_n) \geq g(\lambda)$, whenever $\lambda_n \to \lambda$. In other words, $g(\cdot)$ is lower semicontinuous. In addition, as $g(\lambda) \geq \lambda\delta + \int f(x)d\mu(x)$, we have that $g(\lambda)\to\infty$ if $\lambda\to\infty$. This means that the level sets $\{\lambda \geq 0 : g(\lambda) \leq u\}$ are compact for every u, and therefore, $g(\cdot)$ attains its infimum. In other words, there exists a $\lambda^* \in [0,\infty)$ such that $\bar{I} = \lambda^*\delta + \int \varphi_{\lambda^*}(x)d\mu(x)$. Thus, we conclude that a dual optimizer of the form $(\lambda^*, \varphi_{\lambda^*})$ always exists. Next, if the primal optimizer π^* satisfying $\bar{I} = I(\pi^*)$ also exists, then

$$\int f(y)d\pi^*(x,y) = \lambda^*\delta + \int \varphi_{\lambda^*}(x)d\mu(x),$$

which gives us complementary slackness conditions (10a) and (10b), as a consequence of equality getting enforced in the following series of inequalities:

$$\begin{aligned}\int f(y)d\pi^*(x,y) &= \int (f(y) - \lambda^* c(x,y))d\pi^*(x,y) + \lambda^* \int c(x,y)d\pi^*(x,y)\\ &\leq \int \varphi_{\lambda^*}(x)d\pi^*(x,y) + \lambda^*\int c(x,y)d\pi^*(x,y)\\ &\leq \int \varphi_{\lambda^*}(x)d\mu(x) + \lambda^*\delta.\end{aligned}$$

Alternatively, if the complementary slackness conditions (10a) and (10b) are satisfied by any $\pi^* \in \Phi_{\mu,\delta}$ and $(\lambda^*, \varphi^*_\lambda) \in \Lambda_{c,f}$, then automatically,

$$\begin{aligned}\int f(y)d\pi^*(x,y) &= \int (f(y) - \lambda^* c(x,y))d\pi^*(x,y) + \lambda^* \int c(x,y)d\pi^*(x,y)\\ &= \int \sup_{z\in S}\{f(z) - \lambda^* c(x,z)\}d\pi^*(x,y) + \lambda^*\delta = \int \varphi_{\lambda^*}(x)d\mu(x) + \lambda^*\delta = J(\lambda^*, \varphi_{\lambda^*}),\end{aligned}$$

thus proving that π^* and $(\lambda^*, \varphi_{\lambda^*})$ are the primal and dual optimizers, respectively.

Furthermore, for the uniqueness of the primal optimizer π^*, we argue as follows: for every $x \in S$, if we let $T(x)$ denote the unique maximizer $\arg\max\{f(y) - \lambda^* c(x,y)\}$, then it follows from proposition 7.50(b) of Bertsekas and Shreve [13] that the map $T : (S, \mathcal{U}(S)) \to (S, \mathcal{B}(S))$ is measurable. If (X, Y) is a pair jointly distributed according to π^*, then because of the complementarity slackness condition (10a), we must have that $Y = T(X)$, almost surely. As any primal optimizer must necessarily assign entire probability mass to the set $\{(x,y) : y = T(x)\}$ (because of the complementary slackness condition (10a)), the primal optimizer is unique. This completes the proof of Theorem 1. □

5. On the Existence of a Primal Optimal Transport Plan

Unlike the well-known Kantorovich duality where an optimizer that attains the infimum in (2) exists if the transport cost function $c(\cdot,\cdot)$ is nonnegative and lower semicontinuous (see, e.g., theorem 4.1 in Villani [62]), a primal optimizer π^* satisfying $I(\pi^*) = \bar{I} = \underline{J}$ need not exist for the primal problem, $\bar{I} = \sup\{I(\pi) : \pi \in \Phi_{\mu,\delta}\}$, that we have considered in this paper. The feasible set for the problem (2) in Kantorovich duality, which is the set of all joint distributions $\Pi(\mu,\nu)$ with given μ and ν as marginal distributions, is compact in the weak topology. On the other hand, the primal feasibility set $\Phi_{\mu,\delta}$, for a given μ and $\delta > 0$, that we consider in this paper is not necessarily compact, and the existence of a primal optimizer $\pi^* \in \Phi_{\mu,\delta}$, satisfying $I(\pi^*) = \bar{I}$, is not guaranteed. Example 2 demonstrates a setting where there is no primal optimizer $\pi^* \in \Phi_{\mu,\delta}$ satisfying $I(\pi^*) = \bar{I}$ even if the transport cost $c(\cdot,\cdot)$ is chosen as a metric defined on the Polish space S.

Example 2. Consider $S = \mathbb{R}$. Let $\mu = \delta_{\{0\}}$, the Dirac measure at 0, be the reference measure defined on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Let $f(x) = (1 - \exp(-x))_+$ and $c(x,y) = |x-y|/(1+|x-y|)$ for all $x, y \in \mathbb{R}$. The primal problem of interest is $\bar{I} = \sup\{I(\pi) : \pi \in \Phi_{\mu,\delta}\}$, where $I(\pi) := \int f(y)d\pi(x,y)$, for the choice $\delta = 2$. We first argue that $\bar{I} = 1$ here: as $f(x) \leq 1$ for every $x \in \mathbb{R}$, we have $\bar{I} \le 1$. For $\lambda \ge 0$, recall the definition $\varphi_\lambda(x) := \sup_{y \in \mathbb{R}}\{f(y) - \lambda c(x,y)\}$. As Assumptions (A1) and (A2) hold, an application of Theorem 1(a) results in

$$\begin{aligned}\bar{I} &= \inf_{\lambda \ge 0}\left\{\lambda\delta + \int \varphi_\lambda(x) d\mu(x)\right\} = \inf_{\lambda \ge 0}\{2\lambda + \varphi_\lambda(0)\} \\ &= \inf_{\lambda \ge 0}\left\{2\lambda + \sup_{y \ge 0}\left\{1 - e^{-y} - \lambda \frac{y}{1+y}\right\}\right\} \\ &\ge \sup_{y \ge 0}\inf_{\lambda \ge 0}\left\{1 - e^{-y} + 2\lambda - \lambda\frac{y}{1+y}\right\} = 1.\end{aligned}$$

Since we had already verified that $\bar{I} \le 1$, the above lower bound results in $\bar{I} = 1$, with the infimum being attained at $\lambda^* = 0$. However, as $f(x) < 1$ for every $x \in \mathbb{R}$, it is immediate that $I(\pi) = \int f(y) d\pi(x,y)$ is necessarily smaller than 1 for all $\pi \in \Phi_{\mu,\delta}$. Therefore, there does not exist a primal optimizer $\pi^* \in \Phi_{\mu,\delta}$ such that $I(\pi^*) = \bar{I}$. A careful examination of part (b) of Theorem 1 also results in the same conclusion: a primal optimizer, if it exists, is concentrated on $\{(x,y) : x \in S, y \in \arg\max\{f(y) - \lambda^* c(x,y)\}\}$, which is empty in this example, because $\arg\max\{f(y) - \lambda^* c(x,y)\} = \arg\max_{y \ge 0}\{1 - \exp(-y) - 0 \cdot c(x,y)\} = \varnothing$, for every $x \in S$. □

Recall from Proposition 2 that a primal optimizer always exists whenever the underlying Polish space S is compact. In the absence of compactness of S, Proposition 5 presents additional topological properties (P-Compactness) and (P-USC) under which a primal optimizer exists. Corollary 2, which follows Proposition 5, discusses a simple set of sufficient conditions for which these properties listed in Proposition 5 are easily verified.

5.1. Additional Notation

Define $\Phi'_{\mu,\delta} := \{\pi \in \Phi_{\mu,\delta} : \pi((x,y) \in S \times S : f(x) \le f(y)) = 1\}$. For every $\pi \in \Phi_{\mu,\delta}$, it follows from Lemma B.8 in Appendix B that there exists $\pi' \in \Phi'_{\mu,\delta}$ such that $I(\pi') \ge I(\pi)$, whenever $I(\pi)$ is well defined. As a result,

$$\bar{I} := \sup\{I(\pi) : \pi \in \Phi_{\mu,\delta}\} = \sup\{I(\pi) : \pi \in \Phi'_{\mu,\delta}\}.$$

Throughout this section, we assume that $(\lambda^*, \varphi_{\lambda^*}) \in \Lambda_{c,f}$ is a dual optimal pair satisfying $\bar{I} = \underline{J} = J(\lambda^*, \varphi_{\lambda^*}) < \infty$. As $J(\lambda^*, \varphi_{\lambda^*}) = \lambda^*\delta + \int \varphi_{\lambda^*} d\mu < \infty$, it follows that $\varphi_{\lambda^*}(x) < \infty, \mu$–a.s.

Proposition 5. *Suppose that $\bar{I} = \underline{J}$ is finite, and Assumptions (A1) and (A2) are in force. In addition, suppose that the functions $c(\cdot,\cdot)$ and $f(\cdot)$ are such that the following properties are satisfied.*

Property 1 (P-Compactness). For any $\varepsilon > 0$, there exists a compact $K_\varepsilon \subseteq S$ such that $\mu(K_\varepsilon) > 1 - \varepsilon$, and the closure $\mathrm{cl}(\Gamma_\varepsilon)$ of the set, $\Gamma_\varepsilon := \{(x,y) \in K_\varepsilon \times S : f(y) - \lambda^* c(x,y) \ge \varphi_{\lambda^*}(x) - \gamma\}$, is compact for some $\gamma \in (0,\infty)$.

Property 2 (P-USC). For every collection $\{\pi_n : n \ge 1\} \subseteq \Phi'_{\mu,\delta}$ such that $\pi_n \Rightarrow \pi^*$ for some $\pi^* \in \Phi_{\mu,\delta}$, we have $\overline{\lim}_n I(\pi_n) \le I(\pi^*)$.

Then there exists a primal optimizer $\pi^* \in \Phi_{\mu,\delta}$ satisfying $I(\pi^*) = \bar{I} = \underline{J} = J(\lambda^*, \varphi_{\lambda^*})$.

It is well known that upper semicontinuous functions defined on compact sets attain their supremum. As Property 1 (P-Compactness) and Property 2 (P-USC) respectively enforce a specific type of compactness and upper semicontinuity requirements that are relevant to our setup, we use the suggestive labels (P-Compactness) and (P-USC), respectively, to identify Properties 1 and 2 throughout the rest of this section. Assumptions (A3) and (A4) below, imposing growth conditions on the functions $c(\cdot,\cdot)$ and $f(\cdot)$, provide a simple set of sufficient conditions required to verify the properties (P-Compactness) and (P-USC) stated in Proposition 5.

Assumption 3 (A3). *The level sets $\{(x,y) \in K \times S : c(x,y) \le u\}$ are compact for every $u \in \mathbb{R}$ and compact $K \subseteq S$.*

Assumption 4 (A4). *There exist fixed positive constants M and ε_0 such that for any $\varepsilon \in (0, \varepsilon_0)$, we have $C_\varepsilon \in (0,\infty)$ with the property that $f(y) - f(x) \le M + \varepsilon c(x,y)$ if $(x,y) \in S \times S$ is such that $c(x,y) > C_\varepsilon$.*

As $c(x,x) = 0$ for every $x \in S$, the level sets $\{(x,y) : c(x,y) \le u\}$ are not compact when S is not compact. The requirement imposed in Assumption (A3) is a natural alternative, and it holds, for example, in the following instances:

(a) $S = \mathbb{R}^n$ is the Euclidean space equipped with Euclidean norm $\|\cdot\|$, and $c(x,y) \ge g(\|x-y\|)$ for some $g : \mathbb{R}_+ \to \mathbb{R}_+$ satisfying $g(t) \to \infty$ as $t \to \infty$, and

(b) $S = C[0,T]$ is the space of real-valued continuous functions defined on the interval $[0,T]$ equipped with the supremum norm, and $c(x,y)$ is taken as

$$c(x,y) := |x(0) - y(0)| + \sup_{\substack{s,t \in [0,T] \\ s \neq t}} \frac{|x(t) - y(t)|}{|t-s|^{\alpha}},$$

for some $\alpha \in (0,1]$.

The growth condition imposed in Assumption (A4) is similar to the requirement that the growth rate parameter κ, defined in Gao Kleywegt [31], be equal to 0.

Corollary 2. *Suppose that the functions c and f satisfy Assumptions* (A1)–(A4). *Then, whenever the dual optimal pair* $(\lambda^*, \varphi_{\lambda^*})$ *satisfying* $J(\lambda^*, \varphi_{\lambda^*}) = \underline{J} < \infty$ *is such that* $\lambda^* > 0$, *Properties* (P-Compactness) *and* (P-USC) *are satisfied, and consequently, there exists a primal optimizer* $\pi^* \in \Phi_{\mu,\delta}$ *satisfying* $I(\pi^*) = \bar{I} = \underline{J} = J(\lambda^*, \varphi_{\lambda^*})$.

The proofs of Proposition 5 and Corollary 2 are provided in Section B.5 of the appendix.

Remark 5. In addition to the assumptions made in Corollary 2, suppose that $c(x,y)$ is a convex function in y for every $x \in S$, and f is a strictly concave function. Then, because of Corollary 2, a primal optimizer $\pi^* \in \Phi_{\mu,\delta}$ exists. Furthermore, as the supremum in $\sup_{y \in S}\{f(y) - \lambda^* c(x,y)\}$, is attained at a unique maximizer for every $x \in S$, it follows from the final statement in Theorem 1(b) that the primal optimizer π^* is unique.

6. A Few More Examples

6.1. Applications to Computing General First-Passage Probabilities

Computing probabilities of first passage of a stochastic process into a target set of interest has been one of the central problems in applied probability. The objective of this section is to demonstrate that, similar to the one-dimensional level-crossing example in Section 3, one can compute general worst-case probabilities of first passage into a target set B by simply computing the probability of first passage of the baseline stochastic process into a suitably inflated neighborhood of set B. The goal of such a demonstration is to show that the worst-case first-passage probabilities can be computed with no significant extra effort.

Example 3. Let $R(t) = (R_1(t), R_2(t)) \in \mathbb{R}^2$ model the financial reserve at time t of an insurance firm with two lines of business. Ruin occurs if the reserve process R_t hits a certain ruin set B within time $T > 0$. In the univariate case, the ruin set is usually an interval of the form $(-\infty, 0)$ or its translations (as in Example 1). However, in multivariate cases, the ruin set B can take a variety of shapes based on rules of capital transfers between the different lines of businesses. For example, if capital can be transferred between the two lines without any restrictions, a natural choice is to declare ruin when the total reserve $R_1(t) + R_2(t)$ becomes negative. On the other hand, if no capital transfer is allowed between the two lines, ruin is declared immediately when either $R_1(t)$ or $R_2(t)$ becomes negative. In this example, let us consider the case where the regulatory requirements allow only $\beta \in [0,1]$ fraction of reserve, if positive, to be transferred from one line of business to the other. Such a restriction will result in a ruin set of the form

$$B := \{(x_1, x_2) \in \mathbb{R}^2 : \beta x_1 + x_2 \leq 0\} \cup \{(x_1, x_2) \in \mathbb{R}^2 : \beta x_2 + x_1 \leq 0\}.$$

It is immediately clear that capital transfer is completely unrestricted when $\beta = 1$ and altogether prohibited when $\beta = 0$. The intermediate values of $\beta \in (0,1)$ softly interpolate between these two extreme cases. Of course, one can take the fraction of capital transfer allowed from line 1 to line 2 to be different from that allowed from line 2 to line 1, and various other modifications. However, to keep the discussion simple, we focus on the model described above and refer the readers to Hult and Lindskog [39] for a general specification of ruin models allowing different rules of capital transfers between d lines of businesses.

As in Example 1, we take the space in which the reserve process $(R(t): 0 \leq t \leq T)$ takes values to be $S = D([0,T], \mathbb{R}^2)$, the set of all $\mathbb{R}^2$-valued right continuous functions with left limits defined on the interval $[0,T]$. The space S—again, as in Example 1—is equipped with the standard J_1-metric (see chapter 3 of Whitt [65]), and consequently, the cost function

$$c(x,y) = \inf_{\lambda \in \Lambda} \left(\sup_{t \in [0,T]} \|x(t) - y(\lambda(t))\|_2^2 + \sup_{t \in [0,T]} |\lambda(t) - t| \right), \quad x, y \in S \tag{39}$$

is continuous. Here, $\|x\|_2 = (|x_1|^2 + |x_2|^2|)^{1/2}$ is the standard Euclidean norm for $x = (x_1, x_2)$ in $\mathbb{R}^2$, and Λ is the set of all strictly increasing functions $\lambda : [0,T] \to [0,T]$ such that both λ and λ^{-1} are continuous. Let $\mu \in \mathcal{P}(S)$ denote a baseline probability measure that models the reserve process $R(t)$ in the path space. Given $\delta > 0$, our objective is to characterize the worst-case first-passage probability, $\sup\{P(x(t) \in B \text{ for some } t \in [0,T]) : d_c(\mu, P) \le \delta\}$, of the reserve process hitting the ruin set B. As the set $\{x \in S : x(t) \in B \text{ for some } t \in [0,T]\}$ is not closed, we consider its topological closure,

$$A := \left\{x \in S : \inf_{t\in[0,T]} (\beta x_1(t) + x_2(t)) \le 0\right\} \cup \left\{x \in S : \inf_{t\in[0,T]} (x_1(t) + \beta x_2(t)) \le 0\right\},$$

which, apart from the paths that hit ruin set B, also contains the paths that come arbitrarily close to B without hitting B as a result of the presence of jumps; the fact that A is the desired closure is verified in Lemma B.4 and Corollary B.1 in Appendix B. Furthermore, it is verified in Lemma B.5 in Appendix B that

$$c(x, A) = \frac{1}{1+\beta^2}\left[\inf_{t\in[0,T]} (\beta x_1(t) + x_2(t))^2 \wedge \inf_{t\in[0,T]} (\beta x_2(t) + x_1(t))^2\right].$$

If the reference distribution $\mu(\cdot)$ is such that $h(u) := E_\mu[c(X,A); c(X,A) \le u]$ is continuous, then

$$\sup\{P(x(t) \in B \text{ for some } t \in [0,T]) : d_c(\mu, P) \le \delta\} \le \sup\{P(A) : d_c(\mu, P) \le \delta\} = \mu\left\{x \in S : c(x, A) \le \frac{1}{\lambda^*}\right\}, \tag{40}$$

as an application of Theorem 2; here, as in Section 2.4, $1/\lambda^* = h^{-1}(\delta) := \inf\{u \ge 0 : h(u) \ge \delta\}$. Following the expression for $c(x, A)$ in Lemma B.5, we also have that $\{x \in S : c(x,A) \le 1/\lambda^*\}$ equals

$$\left\{x \in S : \inf_{t\in[0,T]} (\beta x_1(t) + x_2(t)) \le \sqrt{\frac{1+\beta^2}{\lambda^*}}\right\} \cup \left\{x \in S : \inf_{t\in[0,T]} (x_1(t) + \beta x_2(t)) \le \sqrt{\frac{1+\beta^2}{\lambda^*}}\right\}. \tag{41}$$

Next, if we let $c^* = \sqrt{(1+\beta^2)/\lambda^*}$ and

$$A_{(c)} = \left\{x \in S : \inf_{t\in[0,T]} (\beta x_1(t) + x_2(t)) \le c\right\} \cup \left\{x \in S : \inf_{t\in[0,T]} (x_1(t) + \beta x_2(t)) \le c\right\} \tag{42}$$

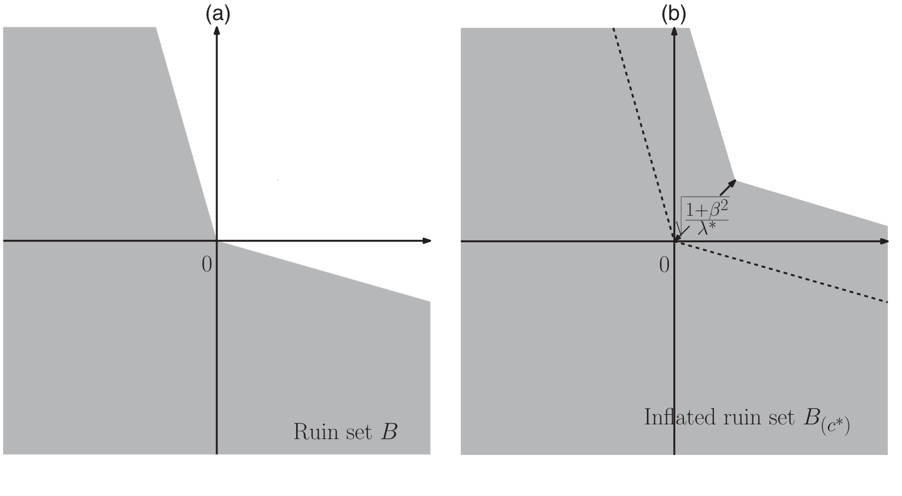
as a family of sets parameterized by $c \in \mathbb{R}$, then it follows from (41) that

$$\sup\{P(A_{(0)}) : d_c(\mu, P) \le \delta\} = \mu(A_{(c^*)}); \tag{43}$$

in other words, the worst-case probability that the reserve process hits or comes arbitrarily close to the ruin set B is simply equal to the probability under reference measure μ that the reserve process hits or comes arbitrarily close to an inflated ruin set $B_{(c^*)}$, where for any $c \in \mathbb{R}$ we define

$$B_{(c)} := \{(x_1, x_2) \in \mathbb{R}^2 : \beta x_1 + x_2 \le c\} \cup \{(x_1, x_2) \in \mathbb{R}^2 : \beta x_2 + x_1 \le c\}.$$

Figure 2. Comparison of computation of ruin under the baseline measure (panel (a)) and worst-case ruin (panel (b)).



This conclusion is very similar to (24) derived for one-dimensional level crossing in Section 3. The original ruin set $B = B_{(0)}$ and the suitably inflated ruin set $B_{(c^*)}$ are respectively shown in panels (a) and (b) of Figure 2.

It may also be useful to note that one may as well choose $S = C([0,T],\mathbb{R}^2),$ the space of continuous functions taking values in $\mathbb{R}^2$, as the underlying space to work with if the modeler decides to restrict the distributional ambiguities to continuous stochastic process; in that case, the first-passage set $\{x \in S : x(t) \in B \text{ for some } t \in [0,T]\}$ itself is closed, and the inequality in (40) holds with equality.

As a final remark, if the modeler believes that model ambiguity is more prevalent in one line of business over other, she can perhaps quantify that effect by instead choosing $\|x\|_2 = (|x_1|^2 + \alpha|x_2|^2)^{1/2}$ for $x = (x_1, x_2) \in \mathbb{R}^2$ in the cost function $c(x,y)$ defined in (39). This would penalize moving probability mass in one direction more than the other and result in the ruin set being inflated asymmetrically along different directions. We identify the study of the effects of various choices of cost functions, the corresponding δ, and the appropriateness of the resulting inflated ruin set for real-world ruin problems as an important direction toward applying the proposed framework in quantitative risk management. For example, suppose that the regulator has interacted with the insurance company, and both the company and the regulator have negotiated a certain level of capital requirement multiple times in the past. Then the insurance company can calibrate δ from these previous interactions with the regulator. That is, find the value of δ that implies that the negotiated capital requirement in a given past interaction is necessary for the bound on ruin probability to be lesser than a fixed (regulatory-driven) acceptance level (for instance, we use 0.01 as the acceptance level for ruin probability in our earlier numerical illustration in this example). An appropriate quantile of these "implied" δ values may be used to choose δ based on risk preference.

6.2. Applications to Ambiguity-Averse Decision Making

In this section, we consider a stochastic optimization problem in the presence of model uncertainty. It has been of immense interest recently to search for distributionally robust optimal decisions that is, to find a decision variable β that solves

$$\mathrm{OPT} = \inf_{\beta \in B} \sup_{P \in \mathcal{P}} E[f(X,\beta)]. \tag{44}$$

Here, f is a performance/risk measure that depends on a random element X and a decision variable β that can be chosen from an action space B. The solution to the above problem minimizes worst-case risk over a family of ambiguous probability measures $\mathcal{P}$. Such an ambiguity-averse optimal choice is also referred to as a distributionally robust choice, because the performance of the chosen decision variable is guaranteed to be better than that obtained by solving irrespective of the model picked from the family $\mathcal{P}$.

There has been a broad range of ambiguity sets $\mathcal{P}$ considered: for examples, refer Delage and Ye [21], Goh and Sim [33], and Wiesemann et al. [66] for moment-based uncertainty sets; Hansen and Sargent [35], Iyengar [41], Nilim and El Ghaoui [51], Lim and Shanthikumar [47], Jain et al. [42], Ben-Tal et al. [11], Wang et al. [64], Jiang and Guan [43], Hu and Hong [38], and Bayraksan and Love [10] for KL divergence and other likelihood-based uncertainty sets; Pflug and Wozabal [52], Wozabal [67], Mohajerin Esfahani and Kuhn [27], Zhao and Guan [69], and Gao and Kleywegt [31] for Wasserstein distance-based neighborhoods; Erdoğan and Iyengar [27] for neighborhoods based on the Prokhorov metric; Bandi et al. [7] and Bandi and Bertsimas [6] for uncertainty sets based on statistical tests; and Ben-Tal et al. [12] and Bertsimas and Sim [14] for a general overview. As most of the works mentioned above assume the random element X to be $\mathbb{R}^d$-valued, it is of our interest in the following example to demonstrate the usefulness of our framework in formulating and solving distributionally robust optimization problems that involve stochastic processes taking values in general Polish spaces as well.

Example 4. We continue with the insurance example considered in Section 3. In practice, as the risk left to the first-line insurer is too large, reinsurance is usually adopted. In proportional reinsurance, one of the popular forms of reinsurance, the insurer agrees with a reinsurer to pay only for a proportion $\beta \in [b,1]$ of the incoming claims for a predetermined $b \in (0,1)$, and the reinsurer pays for the remaining $1-\beta$ fraction of all the claims received. In turn, the reinsurer receives a premium at rate $p_r = (1+\theta)(1-\beta)\nu m_1$ from the insurer. Here, $\theta > \eta$, otherwise, the insurer could make riskless profit by reinsuring the whole portfolio.

The problem we consider here is to find the reinsurance proportion $\beta \in [b,1]$ that minimizes the expected maximum loss that happens within duration T. In the extensive line of research that studies optimal reinsurance proportion, diffusion models have been recommended particularly for tractability reasons (see, e.g., Højgaard and Taksar [37] and Schmidli [57]). As in Example 1, if we take $\sqrt{\nu m_2}B(t) + \nu m_1 t$ to be the diffusion process that approximates the arrival of claims, then

$$L_\beta(t) := p_r t + \beta(\sqrt{\nu m_2}\,B(t) + \nu m_1 t) - pt = \beta\sqrt{\nu m_2}B(t) - (\beta\theta - (\theta - \eta))\nu m_1 t$$

is a suitable model for losses made by the firm. Here, $p_r = (1+\theta)(1-\beta)\nu m_1$ is the rate of payout for the reinsurance contract, and $p = (1+\eta)\nu m_1$ is the rate at which a premium income is received by the insurance firm. The quantity of interest is to determine the reinsurance proportion $\beta \in [b, 1]$ that minimizes the maximum expected losses,

$$L := \inf_{\beta\in[b,1]} E\left[\max_{t\in[0,T]} L_\beta(t)\right]. \tag{45}$$

However, as we saw in Example 1, conclusions based on diffusion approximations can be misleading. Following the practice advocated by the rich literature of robust optimization, we instead find a reinsurance proportion β that performs well against the family of models specified by $\mathcal{P} := \{P : d_c(\mu, P) \le \delta\}$, for a suitable choice of optimal transport cost d_c. In other words, we attempt to solve

$$L' := \inf_{\beta\in[b,1]} \sup_{P\in\mathcal{P}} E_P\left[\sup_{t\in[0,T]} (a_2(\beta)X(t) - a_1(\beta)t)\right], \tag{46}$$

where X, distributed according to measure P, is an element in the space $S = D[0, T]$ equipped with the J_1-topology (see (B.3) for definition of the Skorokhod metric d_{J_1}), $a_2(\beta) := \beta\sqrt{\nu m_2}$, and $a_1(\beta) := (\beta\theta - (\theta - \eta))\nu m_1$. For this example, to define the optimal transport cost d_c, we take the transport cost function as

$$c(x, y) = \inf_{\psi\in\Lambda} (\|x - y\circ\psi\|_\infty + \kappa\|\psi - e\|_\infty)^2 \quad \text{for } x, y \in S, \tag{47}$$

where $\kappa := \max\{\sqrt{\nu/m_2}\cdot \eta m_1/b, 1\}$, $\|z\|_\infty := \sup_{t\in[0,T]}|z(t)|$ for every $z \in S$, $e(t) := t$ is the identity map, and Λ is the set of strictly increasing functions ψ mapping the interval $[0, T]$ onto itself such that both ψ and ψ^{-1} are continuous. Since the Skorokhod metric d_{J_1} satisfies $d_{J_1} \le c \le \kappa d_{J_1}$, we have that $c(\cdot, \cdot)$ is lower semicontinuous. For $x \in S$ and $\beta \in [b, 1]$, if we let

$$f(x, \beta) := \sup_{t\in[0,T]} (a_2(\beta)x(t) - a_1(\beta)t) \quad \text{and} \quad \varphi_{\lambda,\beta}(x) := \sup_{y\in S} \{f(y, \beta) - \lambda c(x, y)\},$$

then because of the application of Theorem 1, we obtain

$$L' = \inf_{\beta\in[b,1]} \sup\{E_P[f(X, \beta)] : d_c(\mu, P) \le \delta\} = \inf_{\beta\in[b,1]} \inf_{\lambda\ge0}\left\{\lambda\delta + \int \varphi_{\lambda,\beta}(x)d\mu(x)\right\}.$$

Recalling that $a_1(\beta) \le \kappa a_2(\beta)$ for any $\beta \in [b, 1]$, it is verified in Lemma B.3 in Appendix B that the inner infimum evaluates simply to

$$E\left[\sup_{t\in[0,T]} (a_2(\beta)B(t) - a_1(\beta)t)\right] + a_2(\beta)\sqrt{\delta}.$$

As a result,

$$L' = \inf_{\beta\in[b,1]} \left\{E\left[\sup_{t\in[0,T]} (a_2(\beta)B(t) - a_1(\beta)t)\right] + a_2(\beta)\sqrt{\delta}\right\}, \tag{48}$$

which is an optimization problem that involves the same effective computational effort as its nonrobust counterpart in (44). For the specific numerical values employed in Example 1, if we also take the additional parameters $\theta = b = 0.3$, the Brownian approximation model evaluates to the optimal choice $\beta = 0.66$ and the corresponding loss $L = 17.63$, whereas its robust counterpart in (46) evaluates to the worst case $L' = 28.86$ for the ambiguity-averse optimal choice $\beta = 0.42$. For a collection of various other examples of using Wasserstein-based ambiguity sets in the context of distributionally robust optimization, refer to Mohajerin Esfahani and Kuhn [49], Zhao and Guan [69], and Gao and Kleywegt [31].

Acknowledgments

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Appendix A. Brownian Embeddings for the Estimation of δ in Example 1

Recall that the definition of optimal transport cost between two probability measures μ and ν, denoted by $d_c(\mu,\nu)$, involves computing the minimum expected cost over all possible couplings between μ and ν (see Section 2.1 for a definition). Although it may not always be possible to identify the optimal coupling (with the lowest expected cost) between μ and ν, one can perhaps employ a "good" coupling to derive an upper bound on $d_c(\mu,\nu)$. In this section, we describe one such coupling, popularly referred as Skorokhod embedding, between the risk process $R(t)$ and its Brownian motion-based diffusion approximation $R_B(t)$ in Example 1. More specifically, we "embed" the compensated compound Poisson process

$$Z(t) = \frac{1}{\sqrt{m_2}}\left(\sum_{i=1}^{N_t} X_i - m_1 t\right)$$

in a Brownian motion $B(t)$ to obtain a coupling between the risk processes $R(t)$ and $R_B(t)$ in order to choose a δ in Example 1. Here, the symbols m_1 and m_2 respectively denote the first and second moments of claim sizes X_i, and N_t is a unit-rate Poisson process. Please refer to Example 1 in Section 3 for a thorough review of the notation. The procedure is data driven in the sense that we do not assume the knowledge of the distribution of claim sizes X_i. Instead, we simply assume access to an oracle that provides independent realizations of claim sizes $\{X_1, X_2, \ldots\}$. Given this access to claim size information, Algorithm A.1 specifies the coupling that embeds the process $Z(t)$ in Brownian motion $B(t)$.

Algorithm A.1. To embed the process $(Z(t): t \geq 0)$ in Brownian motion $(B(t): t \geq 0)$
Given: Brownian motion $B(t)$, moment m_1, and independent realizations of claim sizes $X_1, X_2, \ldots$

Initialize $\tau_0 := 0$ and $\Psi_0 := 0$. For $j \geq 1$, recursively define

$$\tau_{j+1} := \inf\left\{ s \geq \tau_j : \sup_{\tau_j \leq r \leq s} B_r - B_s = X_{j+1} \right\}, \quad \text{and} \quad \Psi_j := \Psi_{j-1} + X_j.$$

Define the auxiliary processes

$$\tilde{S}(t) := \sum_{j>0} \sup_{\tau_j \leq s \leq t} B(s)\mathbf{1}(\tau_j \leq t < \tau_{j+1}) \quad \text{and} \quad \tilde{N}(t) := \sum_{j \geq 0} \Psi_j \mathbf{1}(\tau_j \leq t < \tau_{j+1}).$$

Let $A(t) := \tilde{N}(t) + \tilde{S}(t)$, and identify the time change $\sigma(t) := \inf\{s : A(s) = m_1 t\}$. Next, take the time-changed version $Z(t) := \tilde{S}(\sigma(t))$.

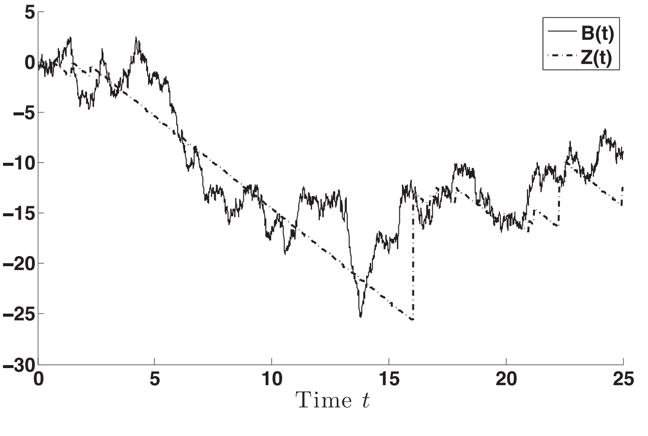
Replace $Z(t)$ by $-Z(t)$ and $B(t)$ by $-B(t)$.

Algorithm A.1 is a brief description of the coupling developed in Khoshnevisan [44], where it is also proved that the process $Z(t)$ output by the construction is indeed the desired compound Poisson process "closely" coupled with the Brownian motion $B(t)$. Figure B.1 shows a typical coupled path output by Algorithm A.1. Several independent replications of such coupled paths are used to simulate the coupled risk processes $R(t)$ and $R_B(t)$ in Example 1, and δ is chosen as prescribed by the confidence intervals (obtained as a result of a straightforward application of central limit theorem) for the empirical average cost of the simulated coupling.

Appendix B. Some Technical Proofs

In this section, we first state and prove all the technical results that are utilized in Examples 1, 3, and 4. Then we prove Lemmas B.6 and B.7, which are technical results used in Section 4 to complete the proof of Theorem 1. Following this, we provide proofs of Lemmas 1 and 2 that are used to prove Theorem 2. Then we conclude this appendix with proofs of Proposition 5

Figure B.1. A coupled path $B(t)$ and $Z(t)$ output by Algorithm A.1.



and Corollary 2 that provide sufficient conditions for existence of a primal optimal transport plan, followed by proofs of Lemma B.8 and Corollary B.2 that add more clarity to the notation

$$\sup\left\{\int f\,dv : d_c(\mu,v)\le\delta\right\} := \sup\left\{\int f\,dv : d_c(\mu,v)\le\delta,\ \int f^{-}dv<\infty\right\} \tag{B.1}$$

adopted while defining the primal problem $\bar{I}$ in Section 2.2.

B.1. Technical Results Used in Examples 1 and 4

For the results (Lemma B.1, B.2, and B.3) stated and proved in this section, let $S = D([0,T],\mathbb{R})$ be equipped with the J_1-topology, and define

$$A_u := \left\{x\in D([0,T],\mathbb{R}) : \sup_{t\in[0,T]} x(t)\ge u\right\},\quad u\in\mathbb{R}.$$

Lemma B.1. *For every $u\in\mathbb{R}$, the set A_u is closed.*

Proof. Pick any $x\notin A_u$, and let $\varepsilon := (u-\sup_{t\in[0,T]}x(t))/2$. As $\sup_{t\in[0,T]}x(t)<u$, ε is strictly positive. Then, for every $y\in S$ such that $d_{J_1}(x,y)<\varepsilon$, we have that

$$\sup_{t\in[0,T]} y(t)\le \sup_{t\in[0,T]} x(t) + d_{J_1}(x,y)\le \sup_{t\in[0,T]} x(t)+\varepsilon,$$

which, in turn, is smaller than u because of the way ε is chosen. Thus, $\{y\in S : d_{J_1}(x,y)<\varepsilon\}$ is a subset of $S\setminus A_u$, and hence $S\setminus A_u$ is open. This automatically means that the set A_u is closed. □

Lemma B.2. *Let $c(x,y)=d_{J_1}(x,y)$, for x,y in $D([0,T],\mathbb{R})$. For any $u\in\mathbb{R}$, $c(x,A_u):=\inf\{c(x,y) : y\in A_u\}$ is given by*

$$c(x,A_u) = u - \sup_{t\in[0,T]} x(t),$$

for every $x\notin A_u$.

Proof. Given $\varepsilon>0$, let $z(t) := x(t)+\inf_{t\in[0,T]}(u-x(t))+\varepsilon$ for all $t\in[0,T]$. As $z\in A_u$, it is immediate from the definition of $c(x,A_u)$ that

$$c(x,A_u)\le d_{J_1}(x,z)\le \sup_{t\in[0,T]}|z(t)-x(t)| = \inf_{t\in[0,T]}(u-x(t))+\varepsilon = u-\sup_{t\in[0,T]}x(t)+\varepsilon. \tag{B.2}$$

Next, let Λ be the set of strictly increasing functions λ mapping the interval $[0,T]$ onto itself, such that both λ and λ^{-1} are continuous. Also, let e be the identity map; that is, $e(t)=t$, for all t in $[0,T]$. Then,

$$d_{J_1}(x,y) := \inf_{\lambda\in\Lambda}\{\|x-y\circ\lambda\|_\infty + \|\lambda - e\|_\infty\}, \tag{B.3}$$

where $\|z\|_\infty = \sup_{t\in[0,T]}|z(t)|$, for any z in S. Since $c(x,y)=d_{J_1}(x,y)$, we have

$$c(x,A_u) = \inf_{y\in A_u}\inf_{\lambda\in\Lambda}\{\|x-y\circ\lambda\|_\infty + \|\lambda-e\|_\infty\}.$$

Next, as $y\circ\lambda\in A_u$ for any $y\in A_u$, it is immediate that one can restrict to $\lambda=e$ without loss of generality, and subsequently, the inner infimum is not necessary. In other words, for any $\varepsilon>0$,

$$c(x,A_u) = \inf_{y\in A_u}\|x-y\|_\infty = \inf_{y\in A_u}\sup_{t\in[0,T]}|x(t)-y(t)| \ge \inf_{y\in A_u}\left(\inf_{t\in[0,T]:y_t>u-\varepsilon} y(t) - \sup_{t\in[0,T]}x(t)\right)\ge u-\sup_{t\in[0,T]}x(t)-\varepsilon.$$

As ε is arbitrary, this observation, when combined with (B.2), concludes the proof of Lemma B.2. □

Lemma B.3. *Let $c(x,y)$ be defined as in (45) for x,y in $D([0,T],\mathbb{R})$. Given nonnegative constants λ, a_1 and a_2, define the functions*

$$f(x) = \sup_{t\in[0,T]}(a_2x(t)-a_1t)\quad\text{and}\quad \varphi_\lambda(x)=\sup_{y\in S}\{f(y)-\lambda c(x,y)\},$$

for every $x \in S$*. If* $a_1/a_2 \leq \kappa$*, then*

$$\inf_{\lambda \geq 0}\left\{\lambda\delta + \int \varphi_\lambda(x)d\mu(x)\right\} = \int f(x)d\mu(x) + a_2\sqrt{\delta}.$$

Proof. Fix any $x \in S$ and $\lambda > 0$. For any positive constant b, as $y(t) = x(t) + b$ is also a member of S, it follows from the definition of $\varphi_\lambda(x)$ that

$$\varphi_\lambda(x) \geq \sup_{\substack{y: y=x+b \\ b\geq 0}}\left\{\sup_{t\in[0,T]}(a_2 y(t) - a_1 t) - \lambda c(x,y)\right\} = \sup_{b\geq 0}\left\{\sup_{t\in[0,T]}(a_2(x(t)+b) - a_1 t) - \lambda b^2\right\}.$$

As the function $a_2 b - \lambda b^2$ is maximized at $b = a_2/2\lambda$, the above lower bound simplifies to

$$\varphi_\lambda(x) \geq f(x) + \frac{a_2^2}{4\lambda}. \tag{B.4}$$

To obtain an upper bound, we first observe from the choice of cost function $c(x,y)$ in (45) that

$$\varphi_\lambda(x) = \sup_{y\in S}\sup_{\psi\in\Lambda}\left\{f(y) - \lambda\big(\|x - y\circ\psi\|_\infty + \kappa\|\psi - e\|_\infty\big)^2\right\},$$

where the quantities Λ and e are defined as in the proof of Lemma B.2. Changing variables $x - y\circ\psi = z$ and $\psi - e = \xi$, we obtain

$$\varphi_\lambda(x) = \sup_{z\in S}\sup_{\xi\in\Lambda}\left\{f\big((x-z)\circ(\xi+e)^{-1}\big) - \lambda(\|z\|_\infty + \kappa\|\xi\|_\infty)^2\right\}.$$

Since $f(x) = \sup_{t\in[0,T]}(a_2 x(t) - a_1 t) = \sup_{t\in[0,T]}(a_2 x(\psi(t)) - a_1\psi(t))$ for any $x \in D[0,T]$ and $\psi \in \Lambda$,

$$f\big((x-z)\circ(\xi+e)^{-1}\big) = \sup_{t\in[0,T]}(a_2(x-z)(t) - a_1(\xi+e)(t)) \leq f(x) + \|a_2 z + a_1\xi\|_\infty.$$

Therefore, $\varphi_\lambda(x) \leq f(x) + \sup_{\psi\in\Lambda}\sup_{z\in S}\{\|a_2 z + a_1\xi\|_\infty - \lambda(\|z\|_\infty + \kappa\,\|\xi\|_\infty)^2\}$. Since $\sup_{x\in\mathbb{R}}\{ax - \lambda(x+b)^2\} = a^2/(4\lambda) - ab$ for any nonnegative a, b, λ, we obtain

$$\varphi_\lambda(x) \leq f(x) + \sup_{\psi\in\Lambda}\sup_{z\in S}\{a_2\|z\|_\infty + a_1\|\xi\|_\infty - \lambda\big(\|z\|_\infty + \kappa\|\xi\|_\infty\big)^2\} \leq f(x) + \frac{a_2^2}{4\lambda} + \sup_{\xi\in\Lambda}\{a_1\|\xi\|_\infty - a_2\kappa\|\xi\|_\infty\} \leq f(x) + \frac{a_2^2}{4\lambda},$$

where the last inequality follows from the assumption that $a_1/a_2 \leq \kappa$. Therefore, $\varphi_\lambda(x) = f(x) + a_2^2/(4\lambda)$. Next, it is a straightforward exercise in calculus to verify that the function $\lambda\delta + \int\varphi_\lambda d\mu$ is minimized at $\lambda^* = a_2/2\sqrt{\delta}$, and subsequently,

$$\inf_{\lambda\geq 0}\left\{\lambda\delta + \int\varphi_\lambda d\mu\right\} = \int f\, d\mu + a_2\sqrt{\delta}.$$

This completes the proof. □

B.2. Proofs of Results Used in Section 6.1

For the results (Lemmas B.4 and B.5 and Corollary B.1) stated and proved in this section, let $S = D([0,T],\mathbb{R}^2)$ be equipped with the J_1-topology induced by the metric

$$d_{J_1}(x,y) = \inf_{\lambda\in\Lambda}\left\{\sup_{t\in[0,T]}\max_{i=1,2}|x_i(t) - y_i(\lambda(t))| + \sup_{t\in[0,T]}|\lambda(t) - t|\right\},$$

where the set Λ, as in Lemma 3, is the set of strictly increasing functions λ mapping the interval $[0,T]$ onto itself, such that both λ and λ^{-1} are continuous.

Lemma B.4. *For a given vector* $a = (a_1, a_2)$ *with* $a_1, a_2 \geq 0$ *and* $a_1 a_2 > 0$*,*

$$\mathrm{cl}\left(\{x\in S : a^T x(t) \leq 0 \text{ for some } t\in[0,T]\}\right) = \left\{x\in S : \inf_{t\in[0,T]} a^T x(t) \leq 0\right\}.$$

Proof. We first show that the set $C := \{x \in S : \inf_{t\in[0,T]} a^T x(t) \le 0\}$ is closed by showing that its complement is open. Given any $x \notin C$, let $\varepsilon := \inf_{t\in[0,T]} a^T x(t) > 0$. Then, for every $y \in S$ such that $d_{J_1}(x,y) < \varepsilon/(2a_1 + 2a_2)$, we have that

$$\inf_{t\in[0,T]} y_i(t) \ge \inf_{t\in[0,T]} x_i(t) - d_{J_1}(x,y) > \inf_{t\in[0,T]} x_i(t) - \frac{\varepsilon}{2(a_1+a_2)},$$

for $i = 1,2$. Furthermore, as a_1, a_2 are nonnegative, we obtain that

$$\inf_{t\in[0,T]} a^T y(t) \ge \inf_{t\in[0,T]} a^T x(t) - (a_1+a_2)\frac{\varepsilon}{2(a_1+a_2)} > 0\,.$$

Thus, for every $x \in S\backslash C$, we can find an $\varepsilon > 0$ such that $\{y \in S : d_{J_1}(x,y) < \varepsilon/(2a_1+2a_2)\}$ is a subset of $S\backslash C$, and hence $S\backslash C$ is open. Therefore, C is closed.

Letting $D := \{x \in S : a^T x(t) \le 0$ for some $t \in [0,T]\}$, the remaining task is to show that for every $\varepsilon > 0$ and $x \in C$, there exists $y \in D$ such that $d_{J_1}(x,y) < \varepsilon$. For any given $\varepsilon > 0$ and $x \in C$, we first see that there exists $t_0 \in [0,T]$ such that $a^T x(t_0) < \varepsilon(a_1+a_2)/2$; such a t_0 exists for every $x \in C$ because $\inf_{t\in[0,T]} a^T x(t) \le 0$. Then $(y(t) = (y_1(t), y_2(t)) : t \in [0,T])$ defined as $y_i(t) = x_i(t) - \varepsilon, i = 1,2$ lies in D because $a^T y(t_0) = a^T x(t_0) - (a_1+a_2)\varepsilon < 0$. Therefore, every x in C is a closure point of D. Therefore, $\mathrm{cl}(D) = C$. □

Corollary B.1. *Let* $B = \{(x_1,x_2) \in \mathbb{R}^2 : \beta x_1 + x_2 \le 0\} \cup \{(x_1,x_2) \in \mathbb{R}^2 : x_1 + \beta x_2 \le 0\}$ *for some* $\beta \ge 0$. *Then* $\mathrm{cl}(\{x \in S : x(t) \in B$ *for some* $t \in [0,T]\})$ *equals* $A_1 \cup A_2$, *where*

$$A_1 := \left\{x \in S : \inf_{t\in[0,T]} (\beta x_1(t) + x_2(t)) \le 0\right\} \quad \text{and } A_2 := \left\{x \in S : \inf_{t\in[0,T]} (x_1(t) + \beta x_2(t)) \le 0\right\}. \tag{B.5}$$

Proof. It follows from Lemma B.4 that $\mathrm{cl}(\{x \in S : \beta x_1(t) + x_2(t) \le 0$ for some $t \in [0,T]\}) = A_1$ and $\mathrm{cl}(\{x \in S : x_1(t) + \beta x_2(t) \le 0$ for some $t \in [0,T]\}) = A_2$. Then the statement to verify follows from the fact that the closure of the union of two sets equals the union of closures of those two sets. □

Lemma B.5. *For* $x, y \in D([0,T], \mathbb{R}^2)$, *let* $c(x,y)$ *be defined as in* (39). *Let* $A = A_1 \cup A_2$, *where* A_1 *and* A_2 *are defined as in* (B.5), *and* $\beta \ge 0$. *Then for* $x \in S\backslash A$,

$$c(x,A) = \frac{1}{1+\beta^2}\left[\inf_{t\in[0,T]} (\beta x_1(t) + x_2(t))^2 \wedge \inf_{t\in[0,T]} (x_1(t) + \beta x_2(t))^2\right],$$

and for $\lambda^* \ge 0, \{x \in S : c(x,A) \le 1/\lambda^*\}$ *equals*

$$\left\{x \in S : \inf_{t\in[0,T]} (\beta x_1(t) + x_2(t)) \le \sqrt{\frac{1+\beta^2}{\lambda^*}}\right\} \cup \left\{x \in S : \inf_{t\in[0,T]} (x_1(t) + \beta x_2(t)) \le \sqrt{\frac{1+\beta^2}{\lambda^*}}\right\}.$$

Proof. Let us focus on determining $c(x,A_1) = \inf_{y\in A_1} c(x,y)$ for $x \notin A_1$. Given $x \notin A_1$ and $\varepsilon > 0$, we first define $\tilde{x}_\varepsilon(t) := x(t) - (b+\varepsilon)u$, where $b := \inf_{t\in[0,T]} (\beta x_1(t) + x_2(t))/\sqrt{1+\beta^2} > 0$ and u is the unit vector (in ℓ_2-norm) along the direction $[\beta, 1]$. As

$$\beta\tilde{x}_{\varepsilon,1}(t) + \tilde{x}_{\varepsilon,2}(t) = \beta x_1(t) + x_2(t) - (b+\varepsilon)\sqrt{1+\beta^2} \le 0 \quad \text{for some } t \le T,$$

we have $\tilde{x}_\varepsilon := (\tilde{x}_{\varepsilon,1}, \tilde{x}_{\varepsilon,2}) \in A_1$. As a result,

$$c(x,A_1) \le c(x,\tilde{x}_\varepsilon) = \|(b+\varepsilon)u\|_2^2 = \left(\inf_{t\in[0,T]} \frac{\beta x_1(t) + x_2(t)}{\sqrt{1+\beta^2}} + \varepsilon\right)^2 \tag{B.6}$$

for $x \notin A_1$. Next, using the same line of reasoning as in the proof of Lemma B.2, one can restrict to time changes $\lambda(t) = t$ in the computation of the lower bound. Then,

$$c(x,A_1) = \inf_{y\in A_1} \sup_{t\in[0,T]} \|x(t) - y(t)\|_2^2 \ge \inf_{y\in A_1} \sup_{t:\beta y_1(t)+y_2(t)\le\varepsilon} \|x(t) - y(t)\|_2^2$$

for any $\varepsilon > 0$. For $z \in \mathbb{R}^2$, if we let $g_\varepsilon(z) = \inf_{\{y\in\mathbb{R}^2 : \beta y_1 + y_2 \le \varepsilon\}} \|z - y\|_2^2$, then

$$c(x,A_1) \ge \inf_{y\in A_1} \sup_{t:\beta y_1(t)+y_2(t)\le\varepsilon} g_\varepsilon(x(t)).$$

Next, for $z = (z_1, z_2)$ such that $\beta z_1 + z_2 > \varepsilon$, observe that $g_\varepsilon(z) = (\beta z_1 + z_2 - \varepsilon)^2/(1+\beta^2)$. Therefore,

$$c(x,A_1) \ge \inf_{y\in A_1} \sup_{t:\beta y_1(t)+y_2(t)\le\varepsilon} \frac{(\beta x_1(t) + x_2(t) - \varepsilon)^2}{1+\beta^2} \ge \inf_{t\in[0,T]} \frac{(\beta x_1(t) + x_2(t) - \varepsilon)^2}{1+\beta^2}$$

for every ε small enough. As ε can be arbitrarily small, combining the upper bound in (B.6) with the above lower bound results in

$$c(x, A_1) = \inf_{t\in[0,T]} \frac{(\beta x_1(t) + x_2(t))^2}{1+\beta^2} \quad \text{for } x \notin A_1.$$

Consequently, $\{x \in S : c(x, A_1) \le 1/\lambda^*\} = A_1 \cup \{x \in S : c(x, A_1) \in (0, 1/\lambda^*)\}$, which is simply

$$\left\{x \in S : \inf_{t\in[0,T]} (\beta x_1(t) + x_2(t)) \le \sqrt{\frac{1+\beta^2}{\lambda^*}}\right\} \quad \text{for } x \notin A_1.$$

Similarly, one can show that $c(x, A_2) = \inf_{t\in[0,T]}(x_1(t) + \beta x_2(t))^2/(1+\beta^2)$ and a similar expression as above for $\{x \in S : c(x, A_2) \le 1/\lambda^*\}$. As $A = A_1 \cup A_2$, it is immediate that $c(x, A) = c(x, A_1) \wedge c(x, A_2)$, and $\{x \in S : c(x, A) \le \frac{1}{\lambda^*}\}$ is the union of two sets $\{x \in S : c(x, A_i) \le 1/\lambda^*\}$, $i = 1, 2$, thus verifying both claims. □

B.3. Technical Results Used in Section 4 to Complete the Proof of Theorem 1

Lemma B.6. *Let $S \times S$ be a Polish space that is compact, and $f : S \to \mathbb{R}$ is upper semicontinuous. Suppose that $\pi \in M(S \times S)$, with the Jordan decomposition $\pi = \pi^+ - \pi^-$ comprising positive measures π^+ and π^-, is such that $\pi^+(A) = 0 < \pi^-(A) < \infty$ for some $A \in \mathscr{B}(S \times S)$. Then,*

$$\inf\left\{\int g d\pi : g \in C_b(S \times S), g(x, y) \ge f(y)\right\} = -\infty.$$

Proof. As any finite Borel measure on a Polish space is regular, there exists a compact set K_ε and an open set O_ε such that $K_\varepsilon \subseteq A \subseteq O_\varepsilon$ and

$$\pi^-(O_\varepsilon) - \varepsilon \le \pi^-(A) \le \pi^-(K_\varepsilon) + \varepsilon, \quad \text{and} \quad \pi^+(O_\varepsilon) \le \varepsilon,$$

for any given $\varepsilon > 0$ (see lemma 18.5 in Aliprantis and Burkinshaw [2]). Furthermore, as $S \times S$ is compact, Urysohn's lemma (see, e.g., theorem 10.8 in Aliprantis and Burkinshaw [2]) guarantees us the existence of a continuous function $h : S \times S \to [0, 1]$ such that $h(x, y) = 1$ for all $x \in K_\varepsilon$ and $h(x, y) = 0$ for all $x \notin O_\varepsilon$. In that case, choosing $\varepsilon < \pi^-(A)/2$, we have $\inf_{n\ge1} \int g_n d\pi = -\infty$ for the sequence of continuous functions $g_n(x, y) = nh(x, y) + \sup_{x\in S} f(x)$. This is because $\sup_{x\in S} f(x) < \infty$ (recall that f is upper semicontinuous and S is compact), and

$$\int h d\pi = \int h d\pi^+ - \int h d\pi^- \le \pi^+(O_\varepsilon) - \pi^-(K_\varepsilon) \le 2\varepsilon - \pi^-(A) < 0$$

when $\varepsilon < \pi^-(A)/2$. As $g_n(x, y) \ge f(y)$ for all x, y in S, it follows that $\inf_{g\in D} \int g d\pi \le \inf_n \int g_n d\pi = -\infty$ as well, whenever there exists A such that $\pi^-(A) > 0$. □

Lemma B.7. *Suppose that Assumptions (A1) and (A2) are in force. Let $(S_n : n \ge 1)$ be an increasing sequence of subsets of S, and let $(\lambda_n : n \ge 1)$ be a real-valued sequence satisfying $\lambda_n \to \lambda^*$, for some $\lambda^* \ge 0$, as $n \to \infty$. Then for any $x \in S$,*

$$\varliminf_n \sup_{y\in S_n} \{f(y) - \lambda_n c(x, y)\} \ge \sup_{y\in\cup_n S_n} \{f(y) - \lambda^* c(x, y)\}.$$

Proof. Pick any $x \in S$. For brevity, let $g(y, \lambda) := f(y) - \lambda c(x, y)$ (hiding the dependence on the fixed choice $x \in S$) and $\overline{\lambda}_m := \sup_{k\ge m} \lambda_k$. The observations that

A. $g(y, \lambda)$ is a nonincreasing function in λ, and
B. $\sup_{y\in S_n} g(y, \lambda) \le \sup_{y\in S_{n+1}} g(y, \lambda)$ for $n \ge 1$,

will be used repeatedly throughout this proof. While observation (B) follows from the fact that $S_n \subseteq S_{n+1}$, observation (A) is true because $c(\cdot, \cdot)$ is nonnegative. The quantity of interest,

$$\varliminf_n \sup_{y\in S_n} \{f(y) - \lambda_n c(x, y)\} = \sup_{n>0} \inf_{m\ge n} \sup_{y\in S_m} g(y, \lambda_m) \ge \sup_{n>0} \inf_{m\ge n} \sup_{y\in S_n} g(y, \lambda_m) \ge \sup_{n>0} \sup_{y\in S_n} \inf_{m\ge n} g(y, \lambda_m),$$

where the first inequality follows from observation (B) and the second inequality from the exchange of inf and sup operations. As $\inf_{m\ge n} g(y, \lambda_m) = f(y) - (\sup_{m\ge n} \lambda_m) c(x, y) = g(y, \overline{\lambda}_n)$, we obtain

$$\varliminf_n \sup_{y\in S_n} \{f(y) - \lambda_n c(x, y)\} \ge \sup_{n>0} \sup_{y\in S_n} g(y, \overline{\lambda}_n) = \sup_{y\in\cup_n S_n} g(y, \lambda^*), \tag{B.7}$$

where the rest of this proof is devoted to justify the last equality in (B.7): as $\sup_{y\in S_n} g(y,\overline{\lambda}_n)$ is nondecreasing in n, it is immediate that $\sup_n \sup_{y\in S_n} g(y,\overline{\lambda}_n)\le \bar{g} := \sup_{y\in\cup_n S_n} g(y,\lambda^*)$. To show that $\sup_n \sup_{y\in S_n} g(y,\overline{\lambda}_n)$ indeed equals $\bar{g}$, we use $\hat{S}$ to denote $\hat{S} := \cup_n S_n$, and we consider distinct cases.

Case 1. When $\bar{g}<\infty$. If $\bar{g}$ is finite, then there exists $y_\varepsilon \in \hat{S}$ such that $g(y_\varepsilon,\lambda^*)\ge \bar{g}-\varepsilon/2$ for any $\varepsilon>0$. Furthermore, as $g(y_\varepsilon,\lambda)$ is continuous in λ, there exists n_0 large enough such that $g(y_\varepsilon,\overline{\lambda}_n)\ge g(y_\varepsilon,\lambda^*)-\varepsilon/2$, and consequently, $g(y_\varepsilon,\overline{\lambda}_n)\ge \bar{g}-\varepsilon$ for all $n\ge n_0$. Therefore, for all $n>n_0$ satisfying $y_\varepsilon\in S_n$, it follows that $\sup_{y\in S_n} g(y,\overline{\lambda}_n)\ge g(y_\varepsilon,\overline{\lambda}_n)\ge \bar{g}-\varepsilon$. Since ε is arbitrary, we obtain $\sup_n\sup_{y\in S_n} g(y,\overline{\lambda}_n)=\bar{g}$ whenever $\bar{g}$ is finite.

Case 2. When $\bar{g}=\infty$. If $\bar{g}=\sup_{y\in\hat{S}} g(y,\lambda^*)=\infty$, then there exists a strictly increasing subsequence $(n_k:k\ge 1)$ of natural numbers such that $\sup_{y\in\hat{S}} g(y,\overline{\lambda}_{n_k})>k$ for all k (this is because, being a pointwise supremum of a family of continuous functions of λ, $\sup_{y\in\hat{S}} g(y,\lambda)$ is a lower semicontinuous function of λ). Next, as $\sup_{y\in\hat{S}} g(y,\overline{\lambda}_{n_k})>k$, there exists a sequence $(y_k:k\ge 1)$ comprising elements of $\hat{S}$ such that $g(y_k,\overline{\lambda}_{n_k})\ge k/2$ for all $k\ge 1$. As $(S_n:n\ge 1)$ is a sequence of sets increasing to $\hat{S}$, one can identify a strictly increasing subsequence $(m_k:k\ge 1)$ of natural numbers such that $y_k\in S_{m_k}$, and consequently, $\sup_{y\in S_{m_k}} g(y,\overline{\lambda}_{n_k})\ge k/2$ for all $k\ge 1$. Next, if we let $l_k = n_k\vee m_k$, then $\overline{\lambda}_{l_k}\le \overline{\lambda}_{n_k}$, $S_{m_k}\subseteq S_{l_k}$, and it follows from observations (A) and (B) that $\sup_{y\in S_{l_k}} g(y,\overline{\lambda}_{l_k})\ge k/2$ as well. This results in a nondecreasing sequence $(l_k:k\ge 1)$ with $l_k\to\infty$ such that $\sup_{y\in S_{l_k}} g(y,\overline{\lambda}_{l_k})\ge k/2$, thus yielding $\sup_n\sup_{y\in S_n} g(y,\overline{\lambda}_n)=\infty=\bar{g}$.

The proof is complete because $\sup_{y\in\cup_n S_n} g(y,\lambda^*)$ in (53) is the desired right-hand side. □

B.4. Technical Results Used in Section 2.4.1 to Complete the Proof of Theorem 2

Proof of Lemma 1. For any fixed $\lambda>0$ and $n>1$, if we let

$$B_n := \left\{(x,y)\in S\times S: \mathbf{1}_A(y)-\lambda c(x,y) > (1-\lambda c(x,A))^+ - n^{-1/2}\right\},$$

then $\pi_n(B_n)\ge 1-n^{-1}/n^{-1/2}=1-n^{-1/2}$, as an application of Markov's inequality utilizing (18). Next, for every $n>1$, given a pair (X_n,Y_n) jointly distributed with law π_n, we define a new jointly distributed pair $(\tilde{X}_n,\tilde{Y}_n)$ as follows:

$$(\tilde{X}_n,\tilde{Y}_n) := \begin{cases}(X_n,Y_n) & \text{if } (X_n,Y_n)\in B_n,\\ (X_n,X_n) & \text{otherwise.}\end{cases}$$

Step 1. We first show that the collection $(\tilde{\pi}_n:n>1)$, where $\tilde{\pi}_n:=\text{Law}(\tilde{X}_n,\tilde{Y}_n)$, satisfies $\lim_n(\int c\,d\tilde{\pi}_n-\int c\,d\pi_n)=0$. For this purpose, we observe that $\tilde{\pi}_n(S\times A)\ge\tilde{\pi}_n((S\times A)\cap B_n)=\pi_n((S\times A)\cap B_n)$, which is at least $\pi_n(S\times A)-\pi_n((S\times S)\backslash B_n)$. Since $\pi_n(B_n)>1-n^{-1/2}$, this results in $\tilde{\pi}_n(S\times A)\ge\pi_n(S\times A)-n^{-1/2}$. Next, it follows from (18) that $\lambda\int c\,d\pi_n$ is at most $\pi_n(S\times A)-\int(1-\lambda c(x,A))^+d\mu(x)+n^{-1}$; we also have that $\lambda\int c\,d\tilde{\pi}_n$ is at least $\tilde{\pi}_n(S\times A)-\int(1-\lambda c(x,A))^+d\mu(x)$, because $\varphi_\lambda(x)=(1-\lambda c(x,A))^+\ge\mathbf{1}_A(y)-\lambda c(x,y)$ for every $x,y\in S$. Combining these observations with $\int c\,d\tilde{\pi}_n=\int_{B_n} c\,d\pi_n+0\le\int c\,d\pi_n$, we obtain

$$0\le\lambda\left(\int c\,d\pi_n-\int c\,d\tilde{\pi}_n\right)\le\pi_n(S\times A)-\int(1-\lambda c(x,A))^+d\mu(x)+n^{-1}-\left(\int(1-\lambda c(x,A))^+d\mu(x)+\tilde{\pi}_n(S\times A)\right)\le 2n^{-1/2},$$

which proves that $\lim_n(\int c\,d\pi_n-\int c\,d\tilde{\pi}_n)=0$ when $\lambda>0$. Therefore, it suffices to consider $\int c\,d\tilde{\pi}_n$ to prove the desired asymptotic bounds on $\int c\,d\pi_n$.

Step 2. For the fixed choice $\lambda>0$ and any $n>1$, define $C_n:=C_n^{(1)}\cup C_n^{(2)}$, where

$$\begin{aligned}C_n^{(1)} &:= \left\{(x,y)\in S\times S:\lambda c(x,A)\le 1+n^{-1/2},\ y\in A,\ \lambda c(x,y)<\lambda c(x,A)+n^{-1/2}\right\}\quad\text{and}\\ C_n^{(2)} &:= \left\{(x,y)\in S\times S:\lambda c(x,A)\in(1-n^{-1/2},\infty),\ y\notin A,\ \lambda c(x,y)<n^{-1/2}\right\}.\end{aligned}$$

Further define $D_n^{(1)}:=\left\{(x,y)\in C_n:\lambda c(x,A)\le 1-n^{-1/2}\right\}$, $D_n^{(2)}:=\left\{(x,y)\in C_n:\lambda c(x,A)>1+n^{-1/2}\right\}$, and $D_n^{(3)}:=C_n\backslash(D_n^{(1)}\cup D_n^{(2)})$. With these definitions, observe that $B_n\subseteq C_n$, and therefore $\pi_n(C_n)\ge 1-n^{-1/2}$. Then $\int c\,d\tilde{\pi}_n=\int_{C_n} c\,d\tilde{\pi}_n+0$ can be bounded from above and below as follows:

$$\int_{D_n^{(1)}} c(x,A)\,d\tilde{\pi}_n(x,y)\le\int c\,d\tilde{\pi}_n\le\int_{D_n^{(1)}\cup D_n^{(3)}}\left(c(x,A)+\frac{1}{\sqrt{n}\,\lambda}\right)d\tilde{\pi}_n(x,y)+\frac{\tilde{\pi}_n(D_n^{(2)})}{\sqrt{n}\lambda}.$$

The next few steps are dedicated to reexpressing the integrals in the upper and lower bounds above to obtain

$$\overline{\lim_n}\int_{D_n^{(1)}\cup D_n^{(3)}}\left(c(x,A)+\frac{1}{\sqrt{n}\,\lambda}\right)d\tilde{\pi}_n(x,y)\le h(1/\lambda)\quad\text{and}\quad\underline{\lim}_n\int_{D_n^{(1)}} c(x,A)\,d\tilde{\pi}_n(x,y)\ge\lim_{v\uparrow 1/\lambda}h(v).\tag{B.8}$$

As every $(x,y) \in D_n^{(1)} \cup D_n^{(3)}$ satisfies $c(x,A) \leq (1+n^{-1/2})/\lambda$, we have

$$\int_{D_n^{(1)} \cup D_n^{(3)}} \left(c(x,A) + \frac{1}{\sqrt{n}\,\lambda}\right) d\tilde{\pi}_n(x,y) \leq \int_{\{x: c(x,A) \leq (1+1/\sqrt{n})/\lambda\} \times S} \left(c(x,A) + \frac{1}{\sqrt{n}\,\lambda}\right) d\tilde{\pi}_n(x,y),$$

which, in turn, converges to $\int_{\{x: c(x,A) \leq 1/\lambda\}} c(x,A) d\mu(x)$ as a consequence of the bounded convergence theorem. This establishes the first inequality in (B.8). Next, as $\{(x,y) \notin D_n^{(1)} : c(x,A) < 1/\lambda\}$ is contained in the union of $(S \times S)\setminus C_n$ and $\{(x,y) \in C_n : (1-1/n)/\lambda < c(x,A) < 1/\lambda\}$, we obtain

$$\begin{aligned}
\lim_{v \uparrow 1/\lambda} h(v) &= \int_{\{x: c(x,A) < 1/\lambda\} \times S} c(x,A) d\tilde{\pi}_n(x,y) \leq \int_{D_n^{(1)}} c(x,A) d\tilde{\pi}_n(x,y) + \frac{1}{\lambda} \tilde{\pi}_n((x,y) \notin D_n^{(1)} : c(x,A) < 1/\lambda) \\
&\leq \int_{D_n^{(1)}} c(x,A) d\tilde{\pi}_n(x,y) + \frac{1}{\lambda}\left(\tilde{\pi}_n((x,y) \in C_n : (1-n^{-1/2})/\lambda \leq c(x,A) < 1/\lambda) + \tilde{\pi}_n((S \times S)\setminus C_n)\right) \\
&\leq \int_{D_n^{(1)}} c(x,A) d\tilde{\pi}_n(x,y) + \frac{1}{\lambda}\left(\mu(x \in S : (1-n^{-1/2})/\lambda \leq c(x,A) < 1/\lambda) + n^{-1/2}\right),
\end{aligned}$$

thus verifying the second inequality in (B.8). This completes the proof of Lemma 1. □

Proof of Lemma 2. When $\lambda = 0$, we have $\varphi_\lambda(x) = (1 - \lambda c(x,A))^+ = \mathbf{1}_{S_A}(x)$, where $S_A := \{x \in S : c(x,A) < \infty\}$. For any $n > 1$, let $B_n := \{(x,y) \in S \times S : \mathbf{1}_A(y) > \mathbf{1}_{S_A}(x) - n^{-1/2}\}$. Then because of Markov's inequality, it follows from (18) that $\pi_n(B_n) > 1 - n^{-1/2}$. Since B_n is the disjoint union of $S_A \times A$ and $(S\setminus S_A) \times S$, we obtain that $\pi_n(S_A \times A) \geq 1 - n^{-1/2} - \mu(S \setminus S_A) = \mu(S_A) - n^{-1/2}$; here, we have also used that $\pi_n(\cdot \times S) = \mu(\cdot)$. As a result, for any $n > 1$,

$$\begin{aligned}
\int c\,d\pi_n &\geq \inf\left\{\int_{S_A \times A} c(x,y) d\pi(x,y) : \pi \in \cup_{\nu \in P(S)} \Pi(\mu,\nu), \pi(S_A \times A) > \mu(S_A) - n^{-1/2}\right\} \\
&\geq \inf\left\{\int_{S_A \times A} c(x,A) d\pi(x,y) : \pi \in \cup_{\nu \in P(S)} \Pi(\mu,\nu), \pi(S_A \times A) > \mu(S_A) - n^{-1/2}\right\},
\end{aligned}$$

where the second inequality follows from the observation that $c(x,y) \geq c(x,A)$ for every (x,y) in $S_A \times A$. Next, if we let $u_n := \inf\{u \geq 0 : \mu(c(x,A) \leq u) > \mu(S_A) - n^{-1/2}\}$, then

$$\int_{S_A \times A} c(x,A) d\pi(x,y) \geq \int_{\{x \in S_A : c(x,A) < u_n\} \times S} c(x,A) d\pi(x,y) = \lim_{v \uparrow u_n} h(v)$$

for any $\pi \in \cup_{\nu \in P(S)} \Pi(\mu,\nu)$ such that $\pi(S_A \times A) > \mu(S_A) - n^{-1/2}$. Therefore, $\int c\,d\pi_n \geq \lim_{v \uparrow u_n} h(v)$ for every $n > 1$. Since u_n increases to the essential supremum of $c(x,A)$ with respect to measure μ as $n \to \infty$, it follows from the monotone convergence theorem that $\underline{\lim}_n\, n \int c\,d\pi_n \geq \lim_{v \uparrow \infty} h(v)$. □

B.5. Proofs for Proposition 5 and Corollary 2 that Provide Sufficient Conditions for Existence of a Primal Optimal Transport Plan

Proof of Proposition 5. As $\bar{I} = \sup\{I(\pi) : \pi \in \Phi'_{\mu,\delta}\}$, consider a collection $\{\pi_n : n \geq 1\} \subseteq \Phi'_{\mu,\delta}$ such that $I(\pi_n) \geq \bar{I} - 1/n$. We first show that Property (P-Compactness) guarantees the existence of a weakly convergent subsequence $(\pi_{n_k} : k \geq 1)$ of $(\pi_n : n \geq 1)$ satisfying $\pi_{n_k} \Rightarrow \pi^*$, for some $\pi^* \in \Phi'_{\mu,\delta}$.

Step 1 (Verifying tightness of $(\pi_n : n \geq 1)$). As $I(\pi_n) \geq \bar{I} - n^{-1}$, it follows from (13) that

$$\int (\varphi_{\lambda^*}(x) - f(y) + \lambda^* c(x,y)) d\pi_n(x,y) \leq \frac{1}{n} \quad \text{and} \quad \lambda^*\left(\delta - \int c\,d\pi_n\right) \leq \frac{1}{n}. \tag{B.9}$$

for every $n \geq 1$. Here, since $J(\lambda^*, \varphi_{\lambda^*}) = \lambda^* \delta + \int \varphi_{\lambda^*} d\mu < \infty$, it follows that $\varphi_{\lambda^*}(x) < \infty$, μ-a.s. For any $a > 0$, Markov's inequality results in

$$\pi_n(\{(x,y) \in S \times S : f(y) - \lambda^* c(x,y) < \varphi_{\lambda^*}(x) - a\}) \leq \frac{1}{a} \int (\varphi_{\lambda^*}(x) - f(y) + \lambda^* c(x,y)) d\pi_n(x,y),$$

which is at most $1/(na)$. Then, given any $\varepsilon > 0$ and for the choice $a = \gamma$ specified in (P-Compactness), it follows from union bound that $\pi_n((S \times S)\setminus \Gamma_{\varepsilon/2})$ is at least

$$\pi_n((S \setminus K_{\varepsilon/2}) \times S) + \pi_n(\{(x,y) \in K_{\varepsilon/2} \times S : f(y) - \lambda^* c(x,y) < \varphi_{\lambda^*}(x) - \gamma\}) \leq \frac{\varepsilon}{2} + \frac{1}{n\gamma},$$

because $\pi_n((S\backslash K_{\varepsilon/2})\times S) = \mu(S\backslash K_{\varepsilon/2}) \le \varepsilon/2$. Consequently, $\pi_n(\Gamma_{\varepsilon/2}) \ge 1-\varepsilon/2 - 1/(n\gamma)$. As the closure of $\Gamma_{\varepsilon/2}$ is compact for the choice of γ in (P-Compactness), we have that $\mathrm{cl}(\Gamma_{\varepsilon/2})$ is a compact subset of $S\times S$ for given $\varepsilon>0$. Therefore, for all $n \ge 2\varepsilon^{-1}\gamma^{-1}$, we obtain that $\pi_n(\mathrm{cl}(\Gamma_\varepsilon)) \ge 1-\varepsilon$ for any given $\varepsilon>0$; thus the collection $\{\pi_n : n\ge 1\}$ is tight.

Then, as a consequence of Prokhorov's theorem, we have a subsequence $(\pi_{n_k} : k\ge 1)$ such that $\pi_{n_k} \Rightarrow \pi^*$ for some $\pi^* \in P(S\times S)$.

Step 2 (Verifying $\pi^* \in \Phi_{\mu,\delta}$). As $\pi_{n_k}(\cdot\times S) = \mu(\cdot)$, we obtain $\pi^*(\cdot \times S) = \mu(\cdot)$ as a consequence of the weak convergence $\pi_{n_k} \Rightarrow \pi^*$ (see that $\int g(x)d\pi^*(x,y) = \lim_k \int g(x) d\pi_n(x,y) = \int g(x)d\mu(x)$ for all $g\in C_b(S)$). Furthermore, as c is a nonnegative lower semicontinuous function, it follows from a version of Fatou's lemma for weakly converging probabilities (see theorem 1.1 in Feinberg et al. [28] and references therein) that $\underline{\lim}_n\, k\int c d\pi_{n_k} \ge \int c d\pi^*$. Therefore, $\pi^* \in \Phi_{\mu,\delta}$.

Step 3 (Verifying optimality of π^*). As Property (P-USC) guarantees that $\overline{\lim}_k I(\pi_{n_k}) \le I(\pi^*)$, we obtain $\bar{I} = \overline{\lim}_k I(\pi_{n_k}) \le I(\pi^*)$, thus yielding $I(\pi^*) = \bar{I}$. The fact that $\bar{I} = \underline{J} = J(\lambda^*, \varphi_{\lambda^*})$ follows from Theorem 1. □

Proof of Corollary 2. We verify the properties (P-Compactness) and (P-USC) in the statement of Proposition 5 to establish the existence of a primal optimizer.

Step 1 (Verification of (P-Compactness)). As any Borel probability measure on a Polish space is tight, there exists a compact $K_\varepsilon \subseteq S$ such that $\mu(K_\varepsilon) \ge 1-\varepsilon$, for any $\varepsilon>0$. Next, for any $\lambda < \lambda^* \wedge \varepsilon_0$, because of Assumption (A4), we have $f(y) - \lambda^* c(x,y) < f(x) + M - (\lambda^*-\lambda)c(x,y)$ whenever $c(x,y) > C_\lambda$ and $f(y) > f(x)$. Without loss of generality, one can take C_λ large enough such that $M - (\lambda^*-\lambda)C_\lambda < -\gamma$ for some $\gamma>0$. Then, since $\varphi_\lambda(x) \ge f(x)$, we have $f(y)-\lambda^* c(x,y) < \varphi_\lambda(x) - \gamma$ for every (x,y) in $\{(x,y): f(y) > f(x), c(x,y) > C_\lambda\}$. As a result,

$$\Gamma_\varepsilon := \{(x,y)\in K_\varepsilon \times S : f(y) - \lambda^* c(x,y) \ge \varphi_{\lambda^*}(x) - \gamma\} \subseteq \{(x,y) : x\in K_\varepsilon,\ c(x,y) \le C_\lambda\}.$$

Since $\{(x,y)\in K_\varepsilon\times S : c(x,y)\le C_\lambda\}$ is compact (because of Assumption (A3)), $\mathrm{cl}(\Gamma_\varepsilon)$ is compact as well, thus verifying the property (P-Compactness).

Step 2 (Verification of (P-USC)). Let $\{\pi_n : n\ge 1\} \subseteq \Phi'_{\mu,\delta}$ be such that $\pi_n \Rightarrow \pi^*$ for some $\pi^* \in \Phi_{\mu,\delta}$. Our objective is to show that

$$\overline{\lim_n}\, I(\pi_n) = \overline{\lim_n} \int f(y) d\pi_n(x,y) \le \int f(y) d\pi^*(x,y) = I(\pi^*). \tag{B.10}$$

While (B.10) follows directly from the properties of weak convergence when f is bounded upper semicontinuous, an additional asymptotic uniform integrability condition that $\overline{\lim}_{K\to\infty} \sup_n \int_{\{(x,y):|f(y)|>K\}} |f(y)| d\pi_n(x,y) = 0$ is sufficient to guarantee (B.10) in the absence of boundedness (see, e.g., corollary 3 in Zapała [68]). To demonstrate this asymptotic uniform integrability property, we proceed as follows: Fix any $\varepsilon\in(0,\varepsilon_0)$. Let K_ε be a compact subset of S such that $\mu(S\backslash K_\varepsilon) < \varepsilon$ and $\int_{S\backslash K_\varepsilon} \varphi_{\lambda^*}(x) d\mu(x) < \varepsilon$ (since $\bar{I} = \underline{J} < \infty$, $\int \varphi_{\lambda^*} d\mu < \infty$ as well). Next, for any $K\in(0,\infty)$, consider the following subsets that partition $F_K := \{(x,y)\in S\times S : f(y) \ge f(x) \vee K\}$:

$$F_{1,K} := \{(x,y)\in F_K : c(x,y) > C_\varepsilon\},\ F_{2,K} := \{(x,y)\in F_K : x\in K_\varepsilon, c(x,y) \le C_\varepsilon\},$$

and $F_{3,K} := F_K \backslash (F_{1,K}\cup F_{2,K})$. Because of growth conditions in Assumption (A4), we have that $\int_{F_{1,K}} f(y) d\pi_n(x,y)$ is at most $\int_{F_{1,K}} (M + \varepsilon c(x,y)) d\pi_n(x,y) \le M\pi_n(F_{1,K}) + \varepsilon\delta$. Next, since $\mathrm{cl}(F_{2,1})$ is compact and f is upper semicontinuous, $a := \sup\{f(y) : (x,y)\in F_{2,1}\}$ is finite. Therefore, $\int_{F_{2,K}} f(y) d\pi_n(x,y) \le a\pi_n(F_{2,K})$. Furthermore, as $f(y) < \varphi_{\lambda^*}(x) + \lambda^* c(x,y)$, we have $\int_{F_{3,K}} f(y) d\pi_n(x,y) \le \int_{S\backslash K_\varepsilon} \varphi_{\lambda^*}(x) d\mu(x) + \lambda^* C_\varepsilon \pi_n(F_{3,K})$. As π_n is concentrated on $\{(x,y) : f(y) \le f(x)\}$ for every n (recall that $\pi_n \in \Phi'_{\mu,\delta}$), we combine the above observations to write

$$\int_{S\times\{|f|>K\}} |f(y) - f(x)| d\pi_n(x,y) \le M\pi_n(F_{1,K}) + \varepsilon\delta + a\pi_n(F_{2,K}) + \varepsilon + C_\varepsilon \pi_n(F_{3,K}). \tag{B.11}$$

Since F_K is contained in the union of $\{(x,y) : f(y) - f(x) > K/2, f(x) < K/2\}$ and $\{(x,y) : f(x) > K/2\}$, it follows (as a consequence of union bound and Markov's inequality) that

$$\pi_n(F_K) \le \pi_n((x,y) : f(y) - f(x) > \mathrm{K}/2) + \pi_n((x,y) : f(x) > K/2) \le \frac{\bar{I} - \int f d\mu}{K/2} + \frac{\int f d\mu}{K/2} = \frac{2\bar{I}}{K},$$

where we have also utilized that $\pi_n(\cdot\times S) = \mu(\cdot)$ and $\int f(y) d\pi_n(x,y) \le \bar{I} < \infty$. Since $F_{i,K} \subseteq F_K$, we observe that $\sup_n \pi_n(F_{i,K}) \le 2\bar{I}/K \to 0$ as $K\to\infty$. Therefore, we obtain from (B.11) that

$$\overline{\lim_{K\to\infty}} \sup_n \int_{S\times\{|f|>K\}} |f(y) - f(x)| d\pi_n(x,y) \le \varepsilon(1+\delta) \tag{B.12}$$

for any $\varepsilon \in (0, \varepsilon_0)$. Since $\overline{\lim}_{K\to\infty} \sup_n \int_{S\times\{|f|>K\}} |f(y)| d\pi_n(x,y)$ does not exceed

$$\overline{\lim_{K\to\infty}} \sup_n \int_{S\times\{|f|>K\}} |f(y) - f(x)| d\pi_n(x,y) + \overline{\lim_{K\to\infty}} \int_{S\times\{|f|>K\}} |f(x)| d\pi_n(x,y),$$

we verify the desired uniform integrability property, because of (B.12) and the dominated convergence theorem applied to the latter term. This verification, in conjunction with the upper semicontinuity of f and weak convergence $\pi_n \Rightarrow \pi^*$, results in (B.10). As all the requirements stated in Proposition 5 are verified, a primal optimizer satisfying $I(\pi^*) = \bar{I}$ exists. ◻

B.6. A Technical Result to Add Clarity to the Notation (B.1)

Lemma B.8. *Suppose that Assumptions* (A1) *and* (A2) *are in force. Then, for every $\pi \in \Phi_{\mu,\delta}$, there exists a $\pi' \in \Phi_{\mu,\delta}$ such that π' is concentrated on $\{(x,y) \in S \times S : f(y) \geq f(x)\}$, along with satisfying*

$$\int f^+(y) d\pi'(x,y) \geq \int f^+(y) d\pi(x,y) \quad \text{and} \quad \int f^-(y) d\pi'(x,y) \leq \int f^-(x) d\mu(x). \tag{B.13}$$

Furthermore, $\int f(y) d\pi'(x,y) \geq \int f(y) d\pi(x,y)$ whenever the integral $\int f(y) d\pi(x,y)$ is well defined.

Proof. Let (X,Y) be jointly distributed according to π. Using (X,Y), let us define a new jointly distributed pair (X', Y'), with joint distribution denoted by π', as follows:

$$X' := X \quad \text{and} \quad Y' := \begin{cases} Y & \text{if } f(X) \leq f(Y), \\ X & \text{otherwise.} \end{cases} \tag{B.14}$$

As $X' = X$, Law $(X') = \mu$. Furthermore, it follows from the definition of (X', Y') in (B.14) that

$$\int c d\pi' = \int_{\{f(x)\leq f(y)\}} c(x,y) d\pi(x,y) + \int_{\{f(x)>f(y)\}} c(x,x) d\pi(x,y) = \int_{\{f(x)\leq f(y)\}} c(x,y) d\pi(x,y),$$

because $c(x,x) = 0$ for every x in S. In addition, as $c(\cdot,\cdot)$ is nonnegative, it follows that $\int c d\pi' \leq \int c d\pi \leq \delta$. Therefore, $\pi \in \Phi_{\mu,\delta}$. Next,

$$\begin{aligned}
\int f^+(y) d\pi'(x,y) &= \int_{\{f(x)\leq f(y)\}} f^+(y) d\pi(x,y) + \int_{\{f(x)>f(y)\}} f^+(x) d\pi(x,y) = \int_{\{f^+(x)\leq f^+(y)\}} f^+(y) d\pi(x,y) + \int_{\{f^+(x)>f^+(y)\}} f^+(x) d\pi(x,y) \\
&\geq \int_{\{f^+(x)\leq f^+(y)\}} f^+(y) d\pi(x,y) + \int_{\{f^+(x)>f^+(y)\}} f^+(y) d\pi(x,y) = \int f^+(y) d\pi(x,y), \quad \text{and} \\
\int f^-(y) d\pi'(x,y) &= \int_{\{f(x)\leq f(y)\}} f^-(y) d\pi(x,y) + \int_{\{f(x)>f(y)\}} f^-(x) d\pi(x,y) = \int_{\{f^-(x)\geq f^-(y)\}} f^-(y) d\pi(x,y) + \int_{\{f^-(x)<f^-(y)\}} f^-(x) d\pi(x,y) \\
&\leq \int_{\{f^-(x)\geq f^-(y)\}} f^-(x) d\pi(x,y) + \int_{\{f^-(x)<f^-(y)\}} f^-(x) d\pi(x,y) = \int f^-(x) d\pi(x,y) = \int f^- d\mu,
\end{aligned}$$

as $\int f^- d\mu < \infty, \int f(y) d\pi'(x,y)$ is always well defined. Furthermore, when $\int f(y) d\pi(x,y)$ is also well defined, it is immediate from the definition of π' that

$$\int f(y) d\pi'(x,y) = \int_{\{f(x)\leq f(y)\}} f(y) d\pi(x,y) + \int_{\{f(x)>f(y)\}} f(x) d\pi(x,y) \geq \int_{\{f(x)\leq f(y)\}} f(y) d\pi(x,y) + \int_{\{f(x)>f(y)\}} f(y) d\pi(x,y) = \int f(y) d\pi(x,y),$$

thus verifying all the claims made in the statement. ◻

Corollary B.2. *Suppose that Assumptions* (A1) *and* (A2) *are in force. Let $\nu \in P(S)$ be such that $\int f^- d\nu = \infty$ and $d_c(\mu,\nu) \leq \delta$ for a given reference probability measure $\mu \in P(S)$. Then there exists $\nu' \in P(S)$ such that $\int f^+ d\nu' \geq \int f^+ d\nu, \int f^- d\nu' < \infty = \int f^- d\nu$, and $d_c(\mu,\nu') \leq \delta$.*

Proof. Let $\pi \in \Pi(\mu,\nu)$ be an optimal coupling that attains the infimum in the definition of $d_c(\mu,\nu)$; in other words, $d_c(\mu,\nu) = \int c d\pi$ (such an optimal coupling π always exists because of the lower semicontinuity of $c(\cdot,\cdot)$; see theorem 4.1 in Villani [62]). Since $d_c(\mu,\nu) \leq \delta$ and $\pi(\cdot \times S) = \mu(\cdot)$, we have that $\pi \in \Phi_{\mu,\delta}$. Then, according to Lemma B.8, there exists $\pi' \in \Phi_{\mu,\delta}$ satisfying (B.12). Consequently, the marginal distribution defined by $\nu'(\cdot) := \pi'(S \times \cdot)$ satisfies $d_c(\mu,\nu') \leq \delta$, along with $\int f^+ d\nu' \geq \int f^+ d\nu$ and $\int f^- d\nu' \leq \int f^- d\mu < \infty = \int f^- d\nu$. ◻

Endnotes

[1] See, for example, example 3.100 and theorem 5.26 in Aliprantis and Border [1]. Also, note that another version of Gao and Kleywegt [31] was posted in the arXiv repository several months after an initial version of this paper; we have not examined the revised proof carefully.

[2] This is because, under the assumption that $f \in L^1(d\mu)$, for every probability measure ν such that $d_c(\mu, \nu) \leq \delta$ and $\int f^- d\nu = +\infty$, one can identify a probability measure ν' satisfying $d_c(\mu, \nu') \leq \delta, \int f^+ d\nu' \geq \int f^+ d\nu$, and $\int f^- d\nu' < +\infty = \int f^- d\nu$; see Corollary B.2 in Appendix B for a simple construction of such a measure ν' from measure ν. Consequently, when computing the supremum of $\int f d\nu$, it is meaningful to restrict our attention to probability measures ν satisfying $\int f^- d\nu < +\infty$ and interpret $\sup\{\int f d\nu : d_c(\mu, \nu) \leq \delta\}$ as $\sup\{\int f d\nu : d_c(\mu, \nu) \leq \delta, \int f^- d\nu < +\infty\}$. The convention $+\infty - \infty := +\infty$, in this context, simply facilitates the above interpretation.

[3] As $\sup\{f(y) - \lambda c(x, y)\} \geq f(x)$, there is no ambiguity, such as the form $+\infty - \infty$ in $\int \sup_{y \in S}\{f(y) - \lambda c(x, y)\} d\mu(x)$.

[4] In the literature of optimal transport, it is common to refer joint probability measures alternatively as transport plans and the integral $\int c d\pi$ as the cost of transport plan π. We follow this convention to allow for ease of interpretation.

[5] The assumption that $\mathrm{Proj}_A(x) \neq \varnothing$ holds, for example, when A is compact and nonempty (as c is lower semicontinuous) or when $c(x, \cdot)$ has compact sublevel sets for each x (recall that A is a closed set in the discussion).

[6] Let $f : X \to \mathbb{R}$ be an upper semicontinuous function, bounded from above, defined on a Polish space X. If $d(\cdot, \cdot)$ is a function that metrizes the Polish space X, then, for instance, the choice $f_n(x) = \sup_{y \in X}\{f(y) - nd(x, y)\}$ is continuous and satisfies $f_n \downarrow f$ pointwise.

[7] For instance, one can choose $c_n(x, y) = \inf_{\tilde{x}, \tilde{y} \in S}\{c(\tilde{x}, \tilde{y}) + nd((x, y), (\tilde{x}, \tilde{y}))\}$, where the function d metrizes the Polish space $S \times S$. Then $c_n(\cdot, \cdot)$ is continuous and satisfies $c_n(x, x) = 0$ for every $x \in S$.

[8] See, for example, chapter 7 of Bertsekas and Shreve [13] for an introduction to analytic subsets and the Jankov–von Neumann measurable selection theorem.

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